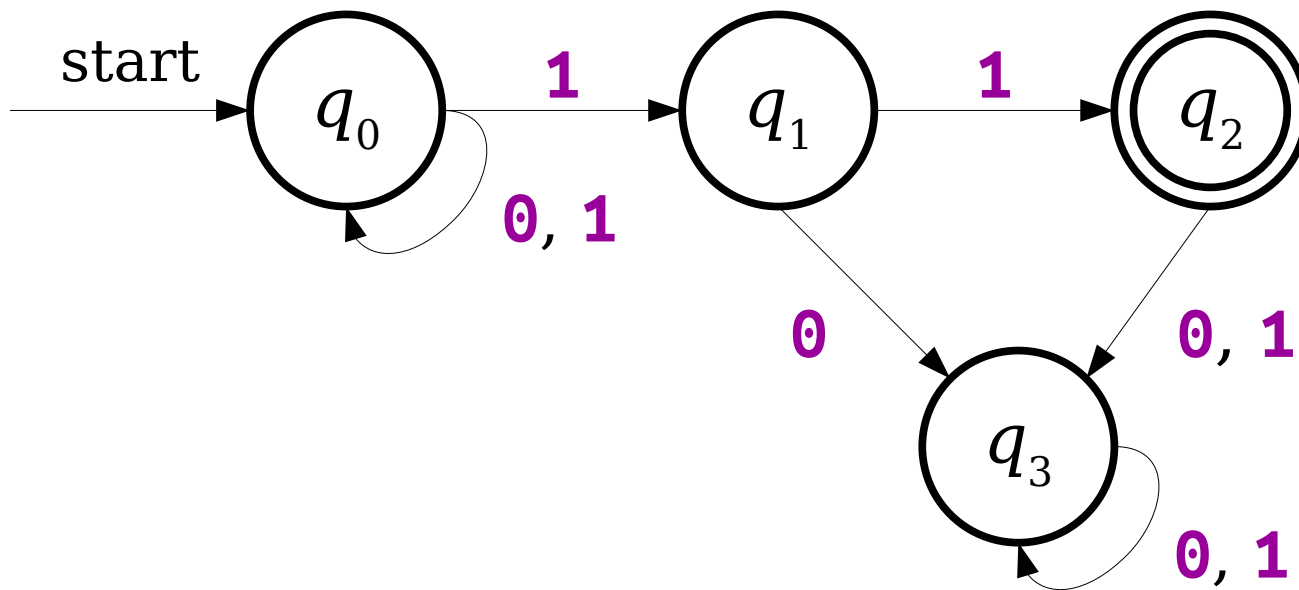


NFAS

The Motivation



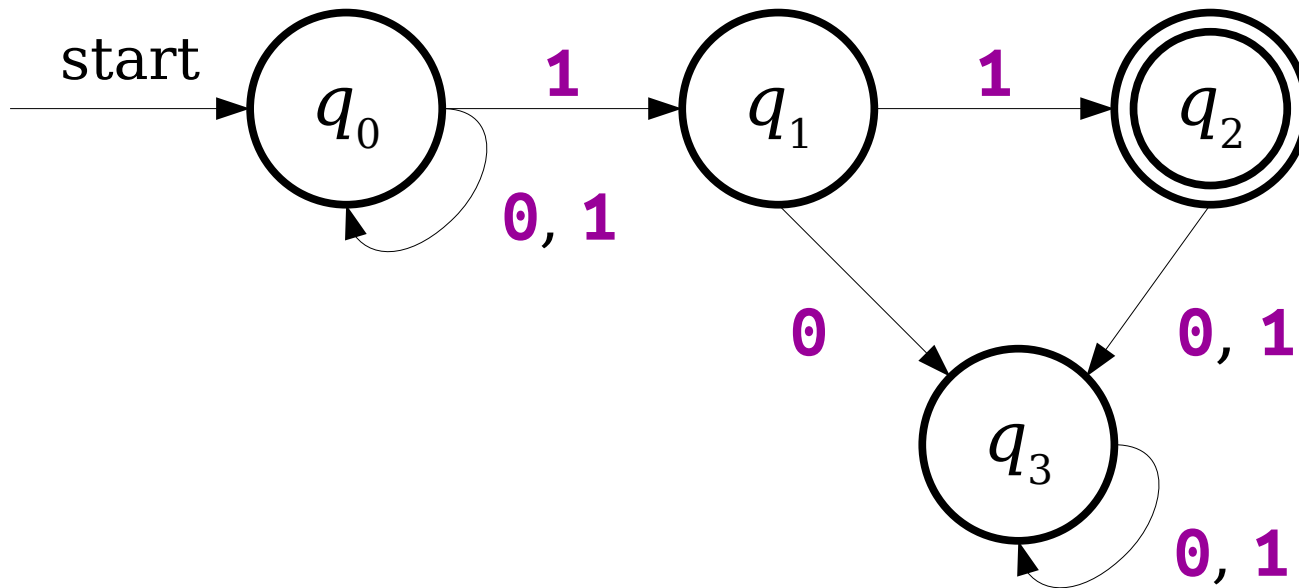
NFAs

- An **NFA** is a
 - **N**ondeterministic
 - **F**inite
 - **A**utomaton
- Structurally similar to a DFA, but represents a fundamental shift in how we'll think about computation.

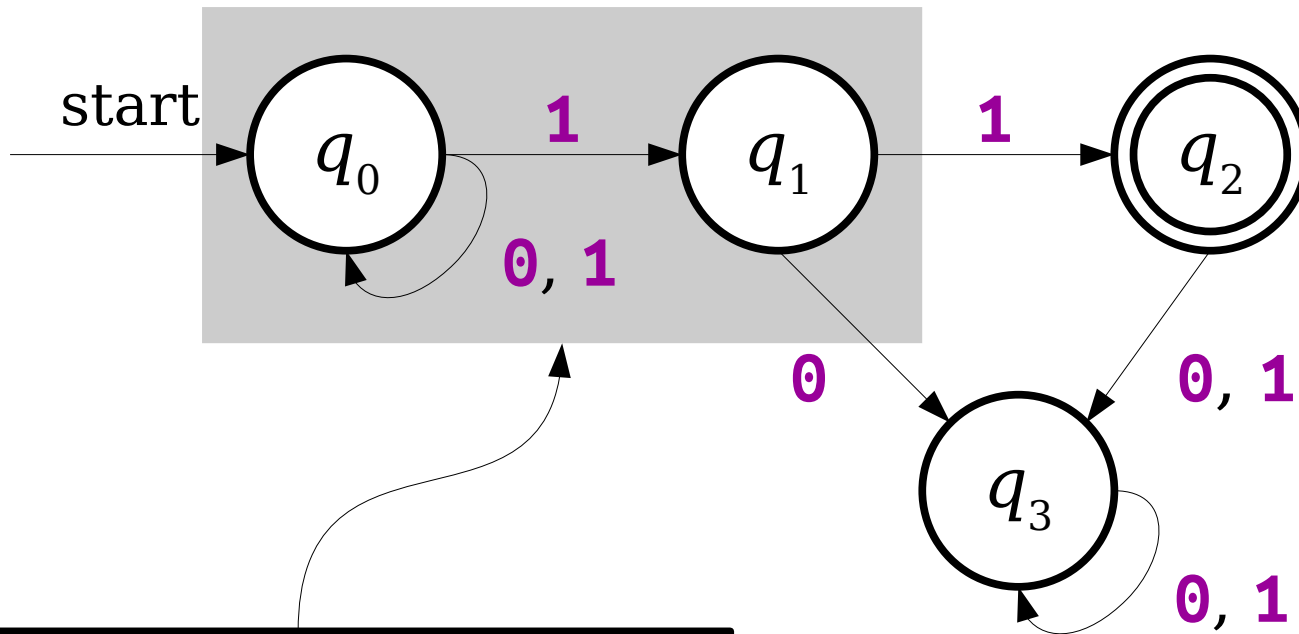
(Non)determinism

- A model of computation is **deterministic** if at every point in the computation, there is exactly one choice that can make.
 - The machine accepts if that series of choices leads to an accepting state.
- A model of computation is **nondeterministic** if the computing machine has a finite number of choices available to make at each point, possibly including zero.
- The machine accepts if **any** series of choices leads to an accepting state.
 - (This sort of nondeterminism is technically called **existential nondeterminism**, the most philosophical-sounding term we'll introduce all quarter.)

A Simple NFA

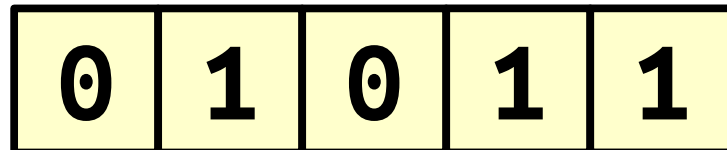
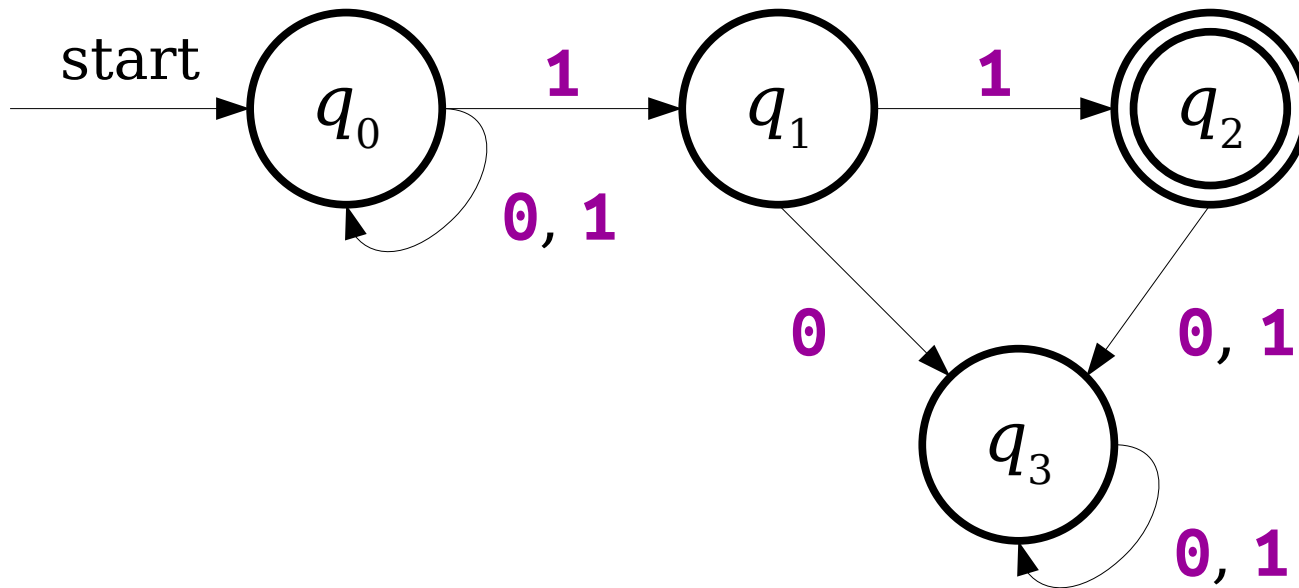


A Simple NFA

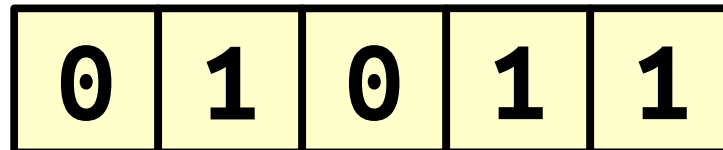
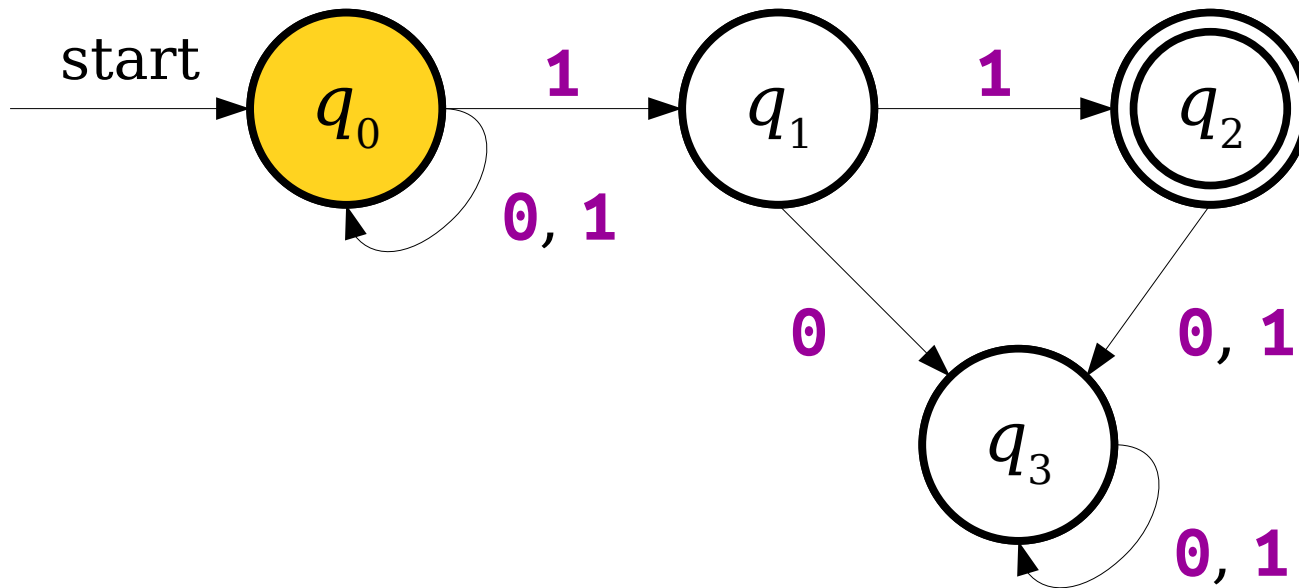


q_0 has two transitions
defined on 1!

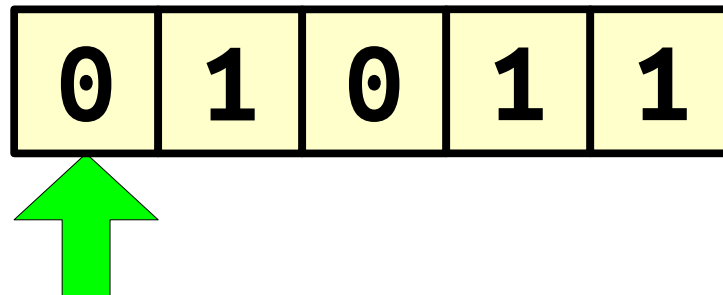
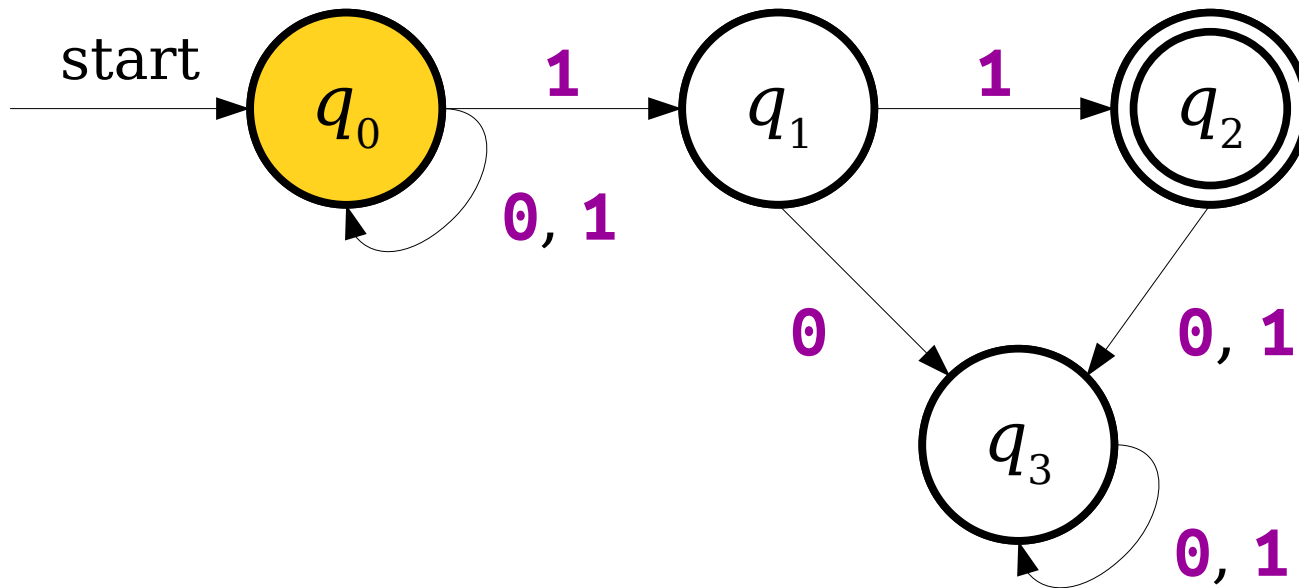
A Simple NFA



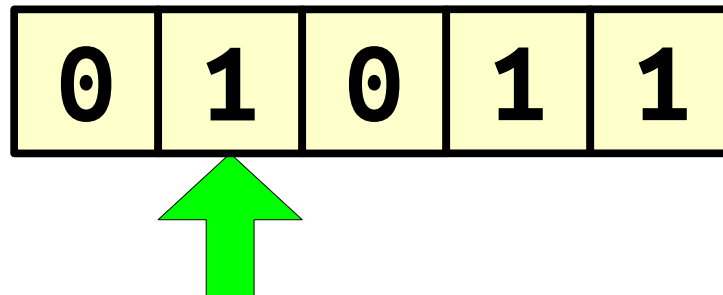
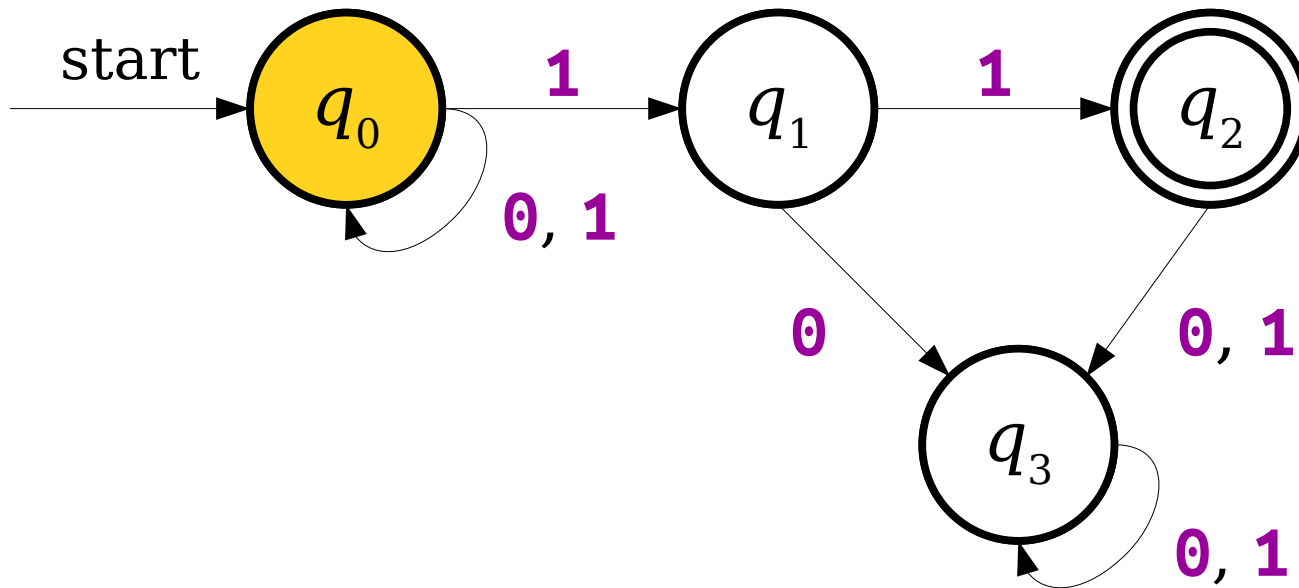
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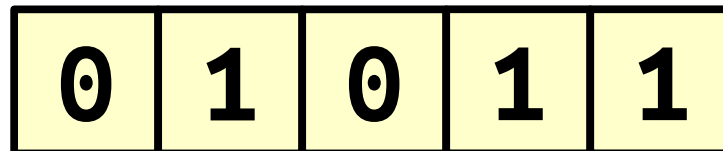
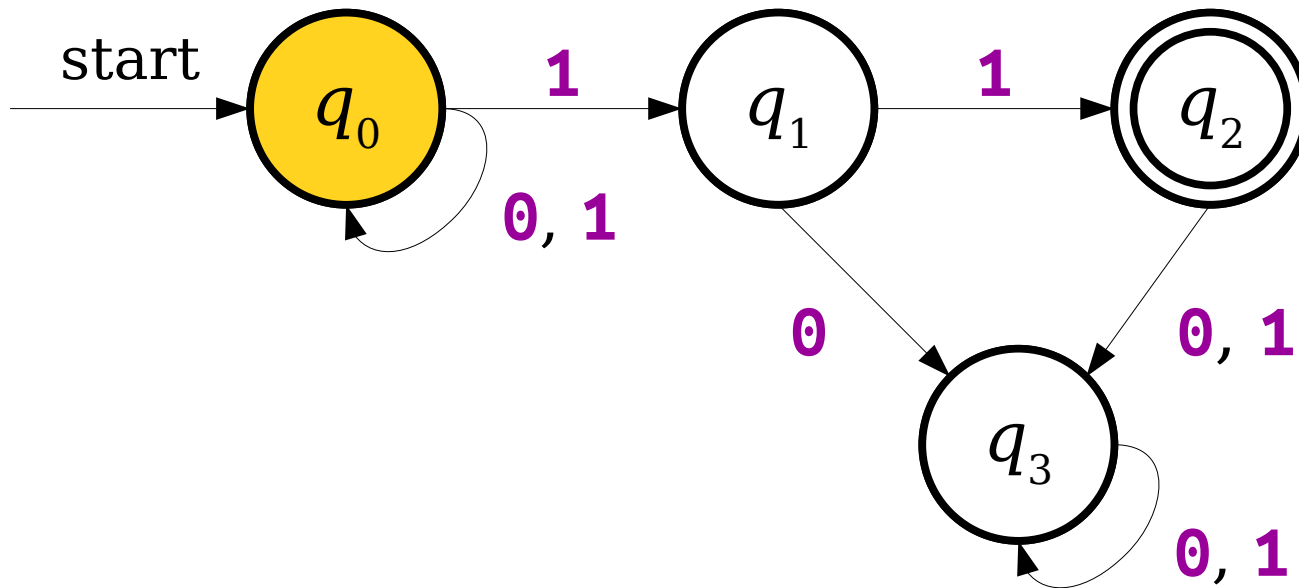
A Simple NFA



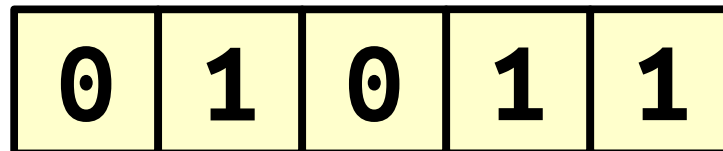
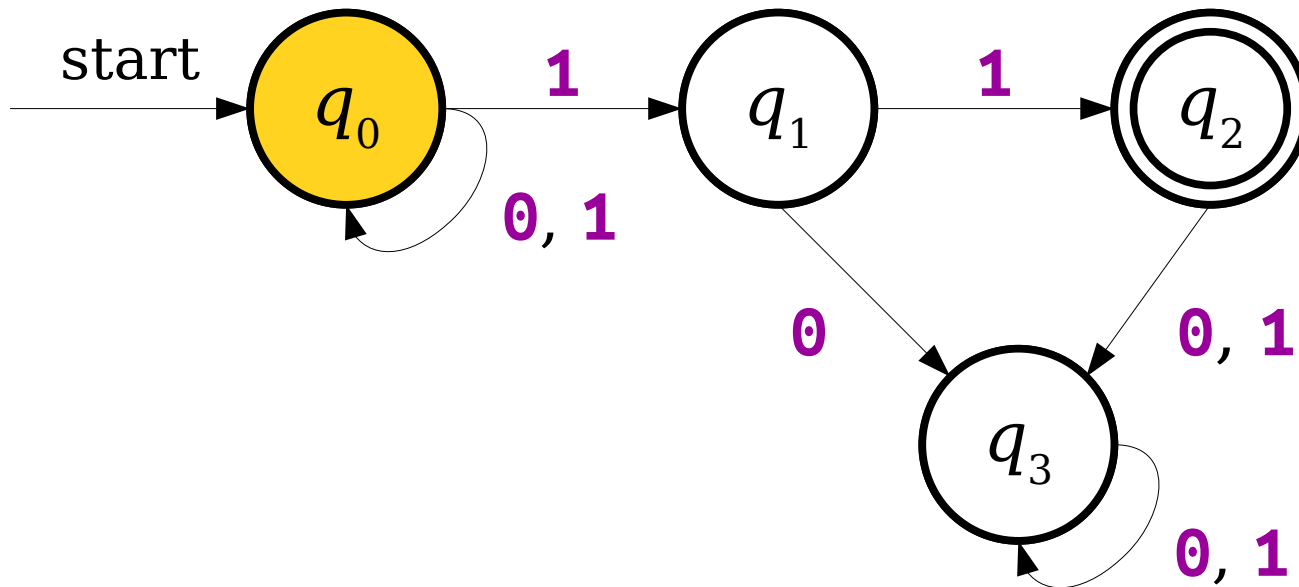
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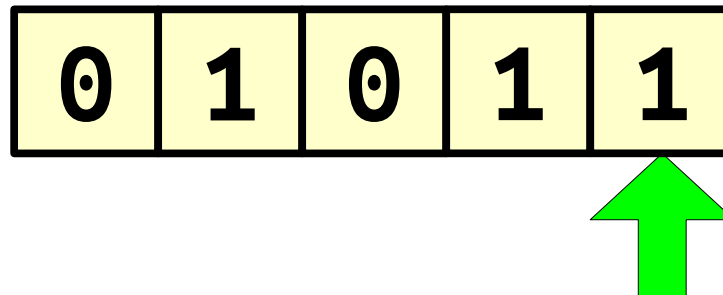
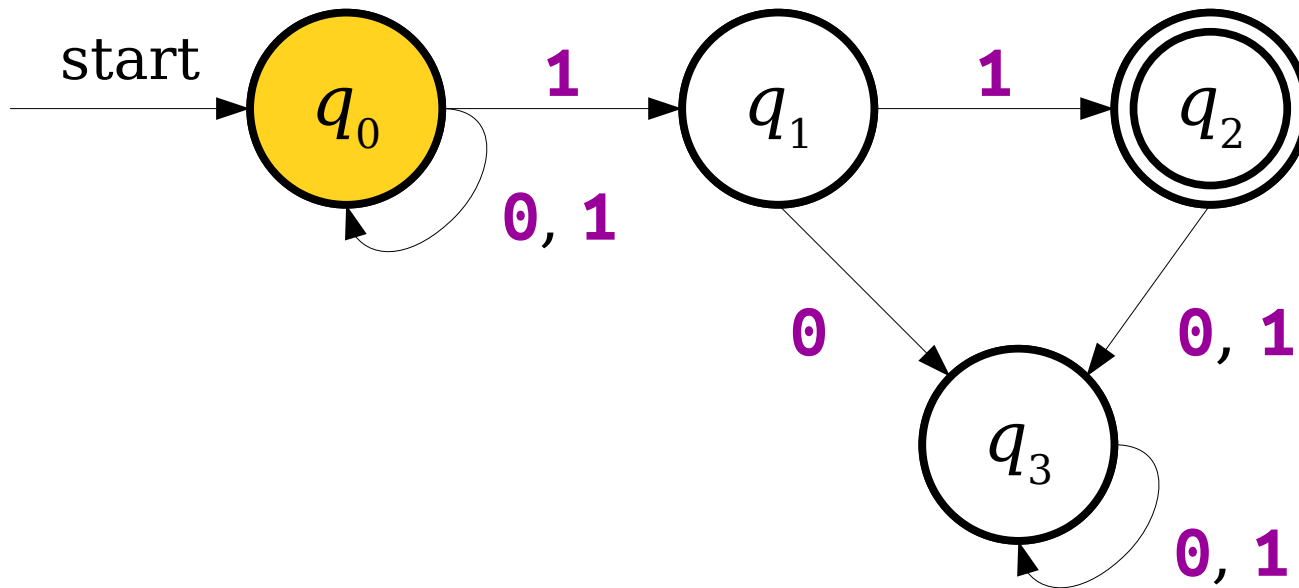
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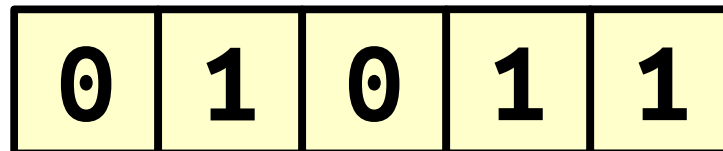
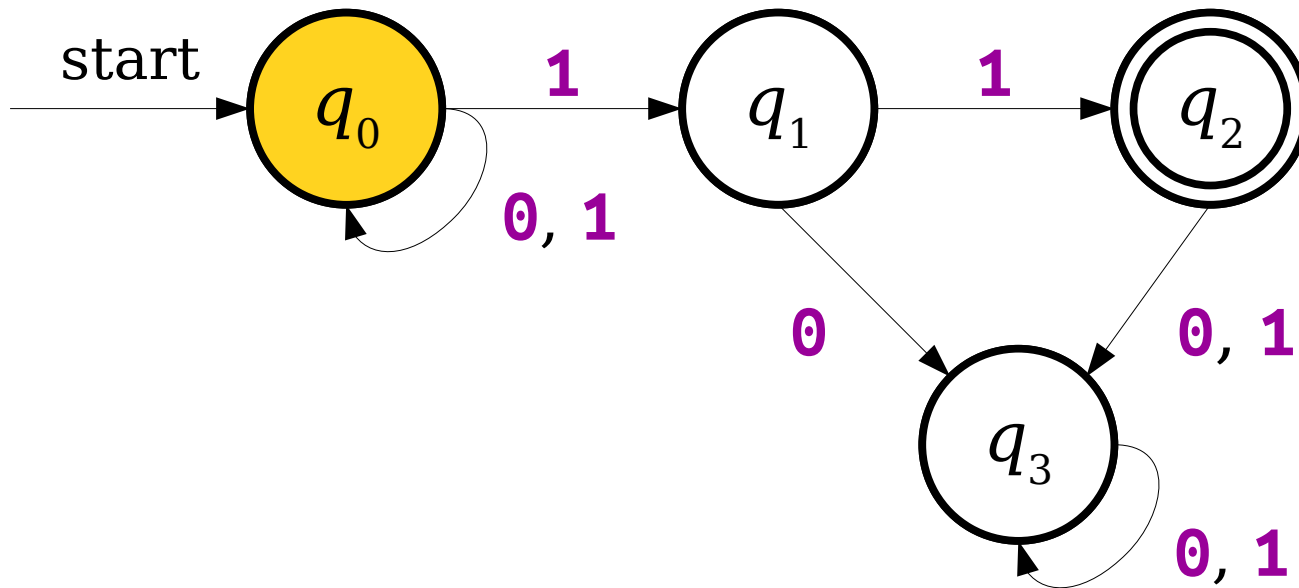
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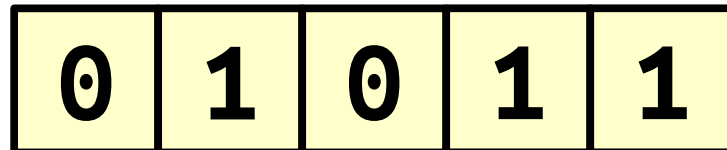
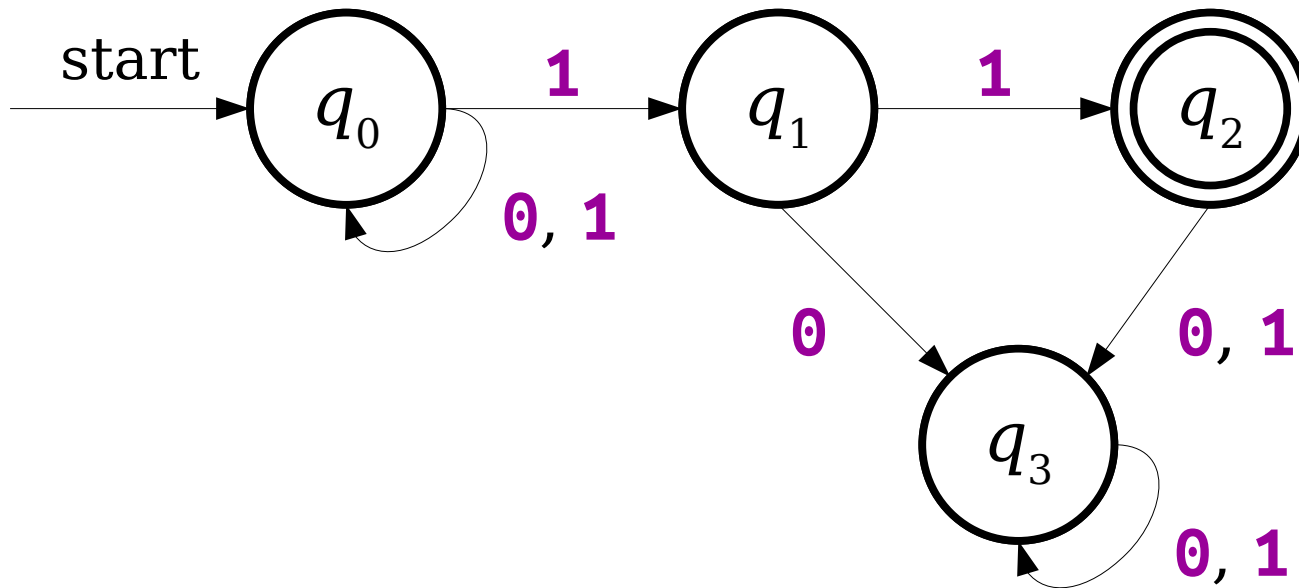
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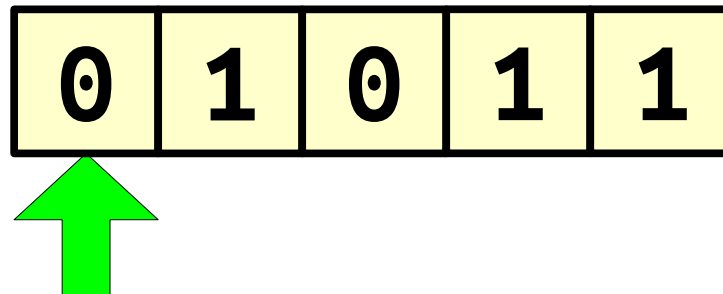
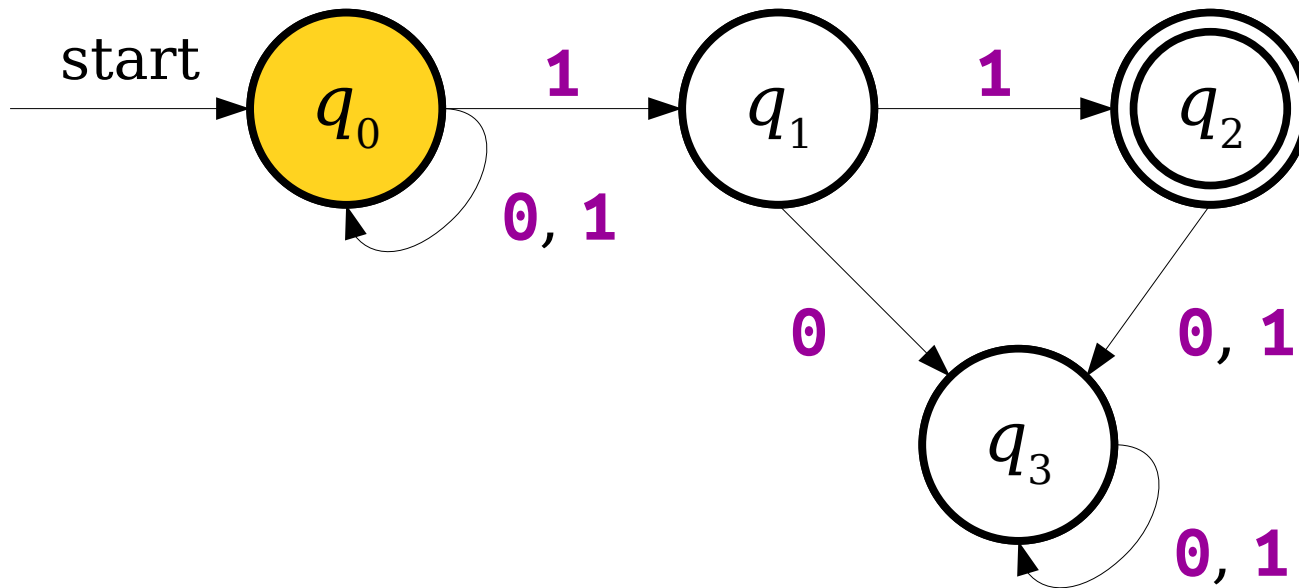
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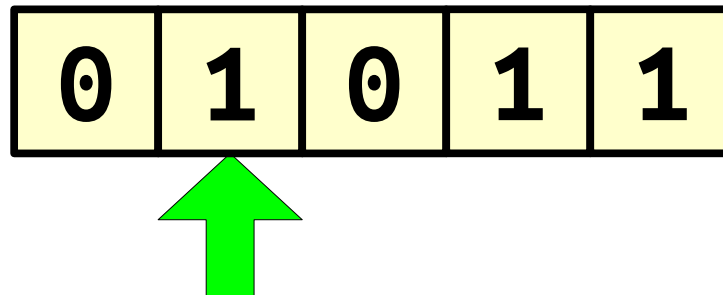
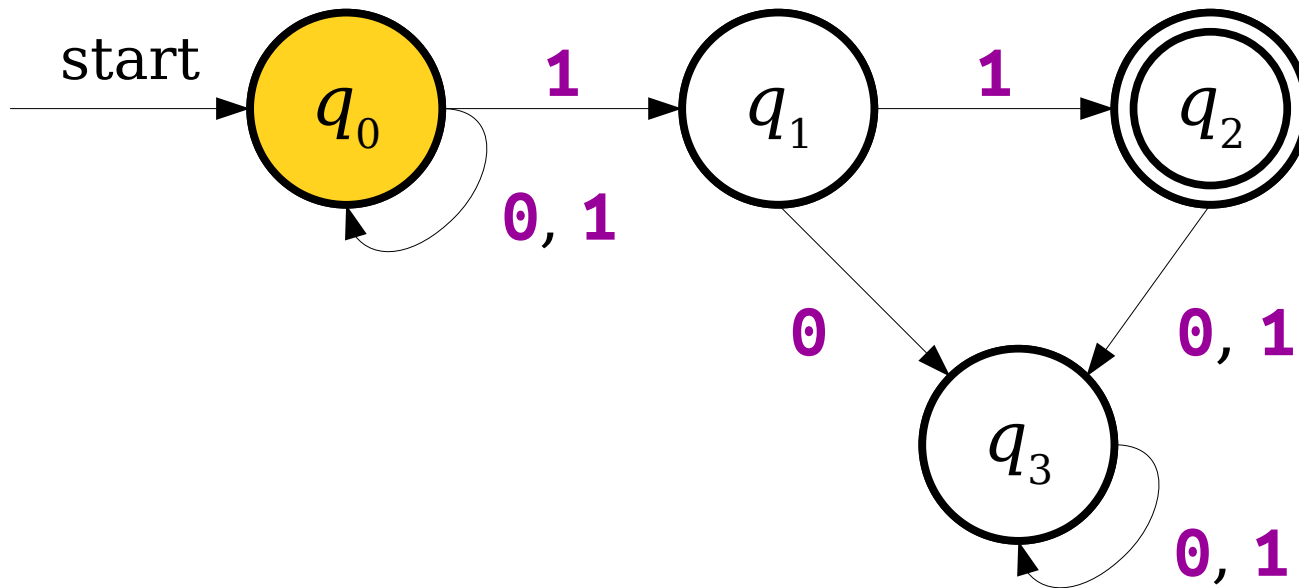
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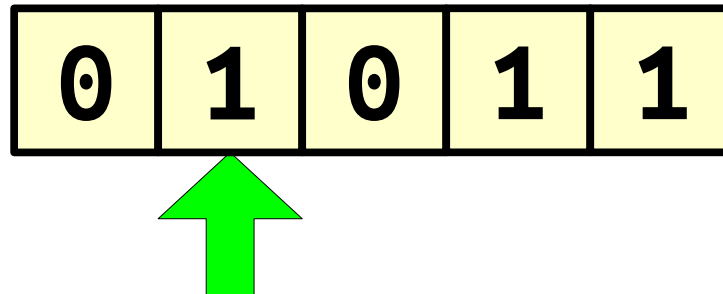
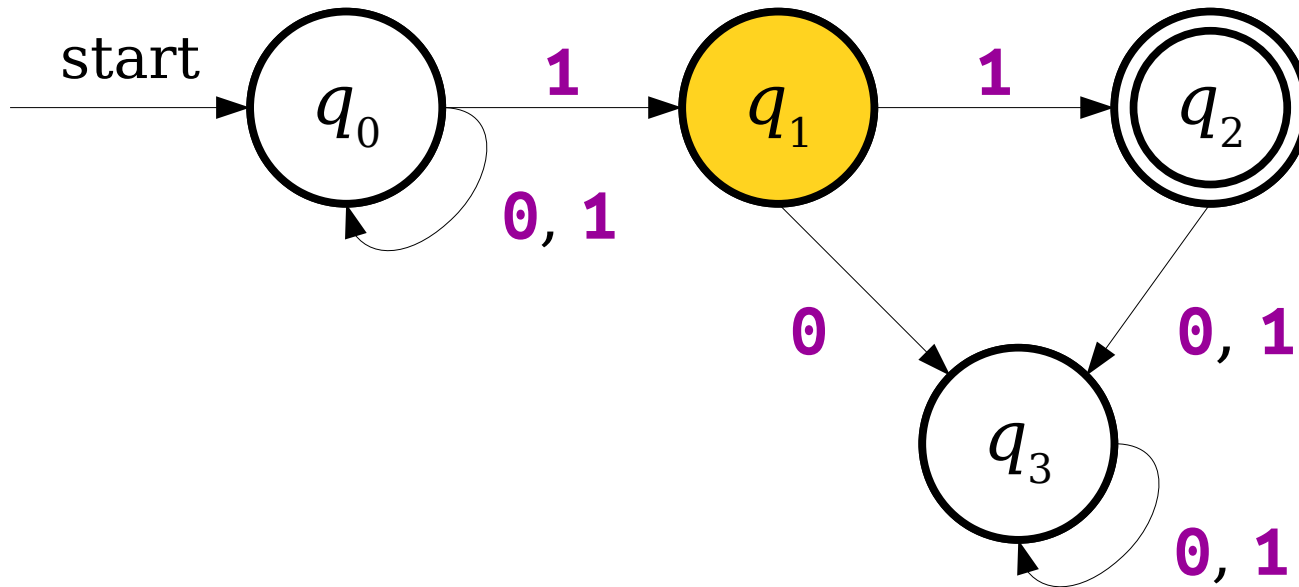
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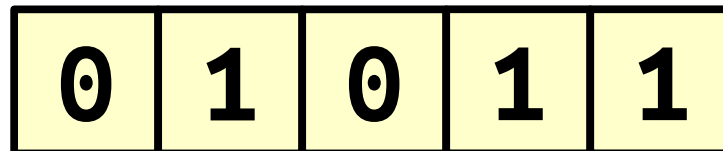
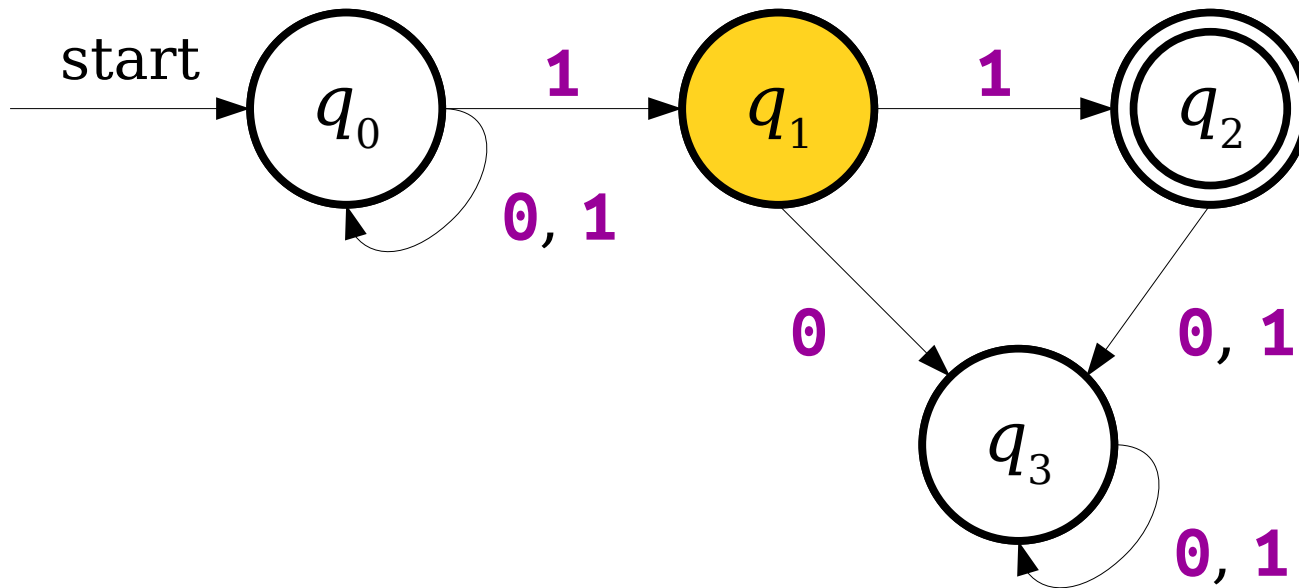
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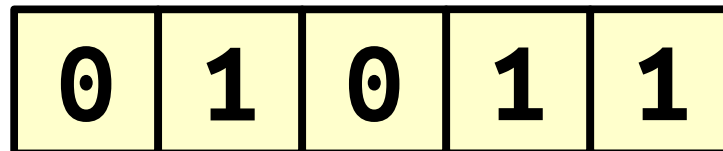
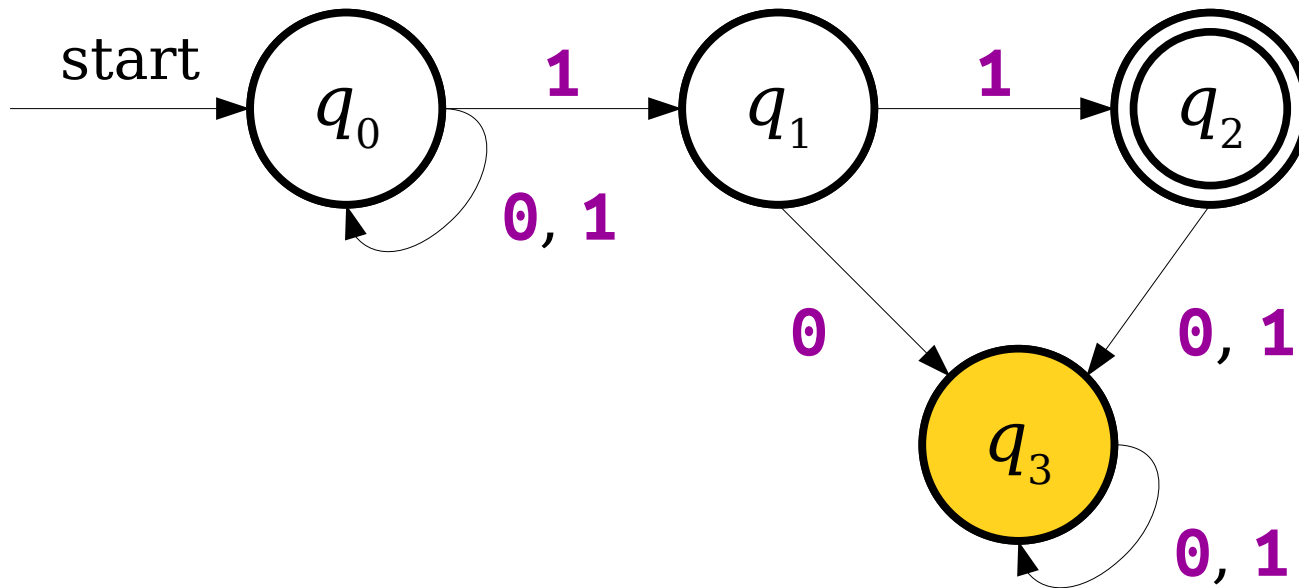
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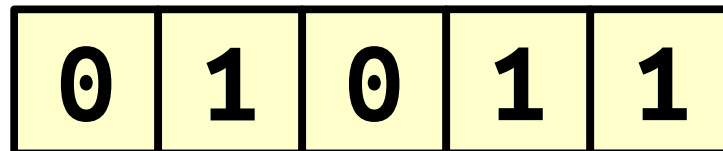
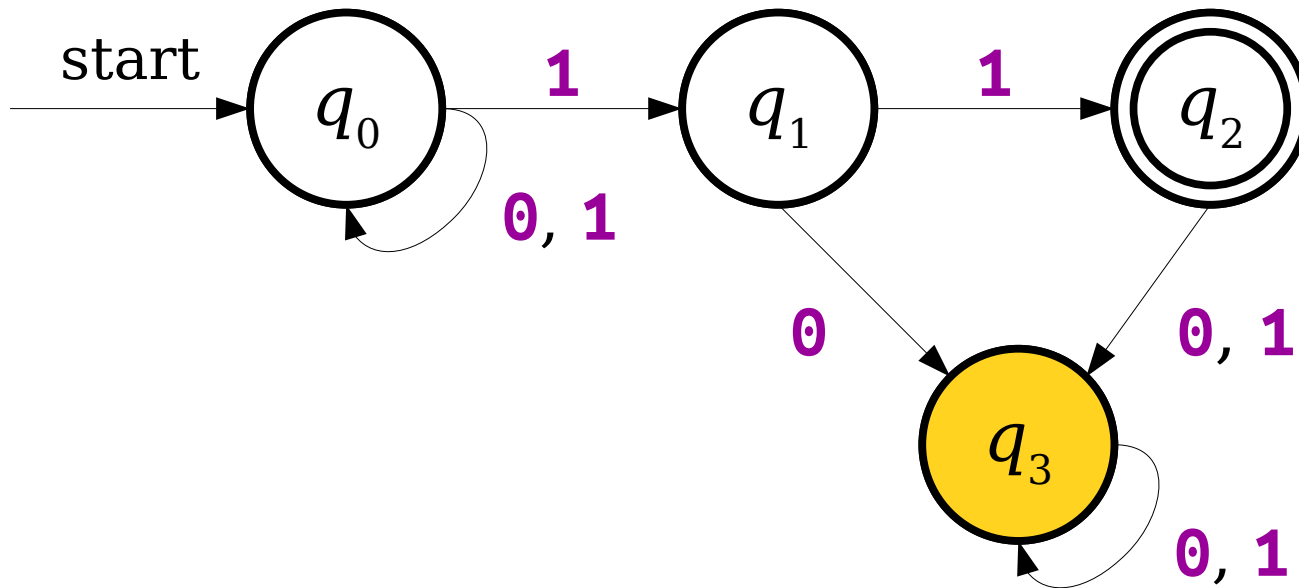
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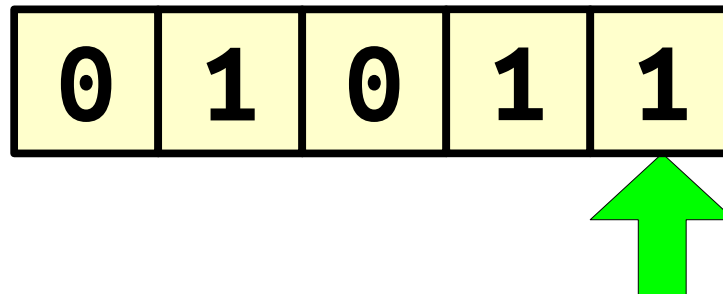
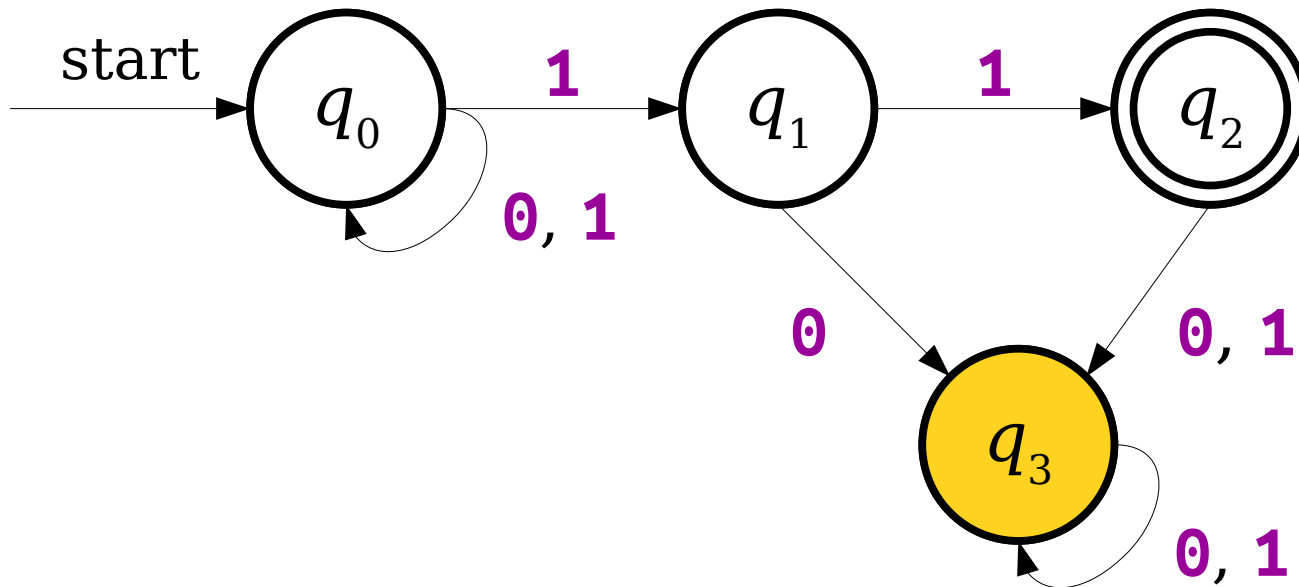
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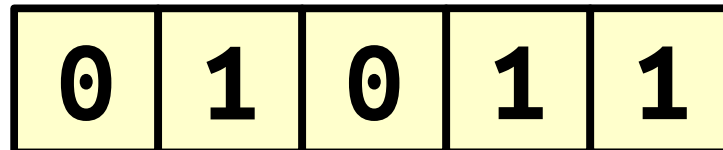
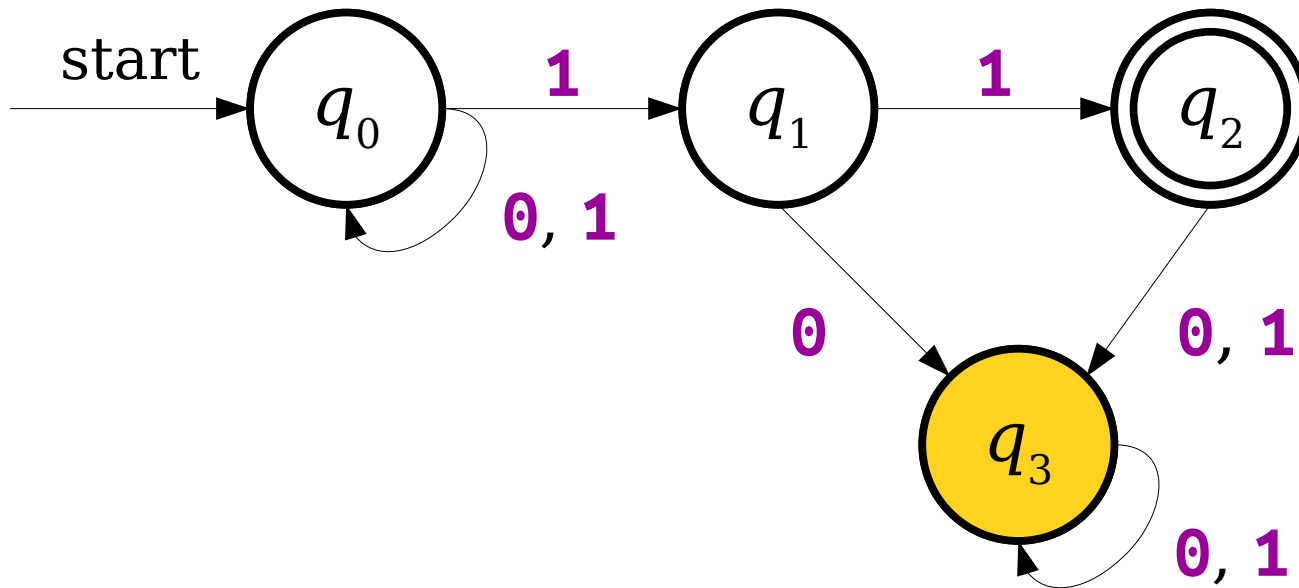
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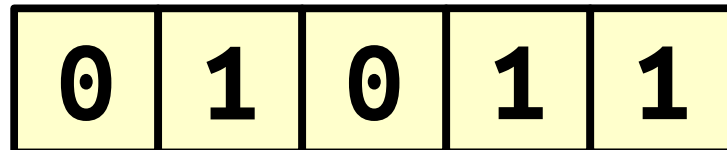
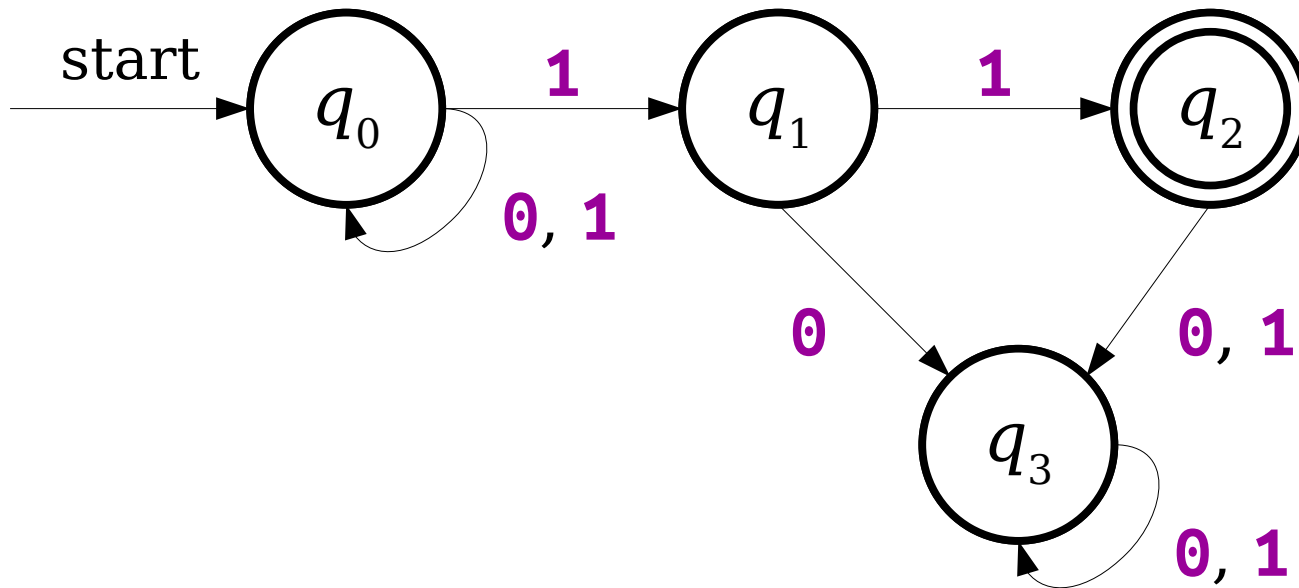
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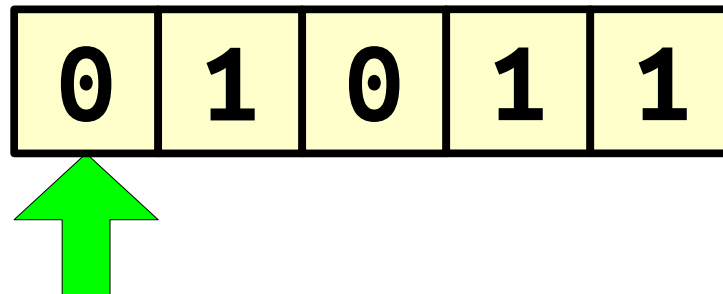
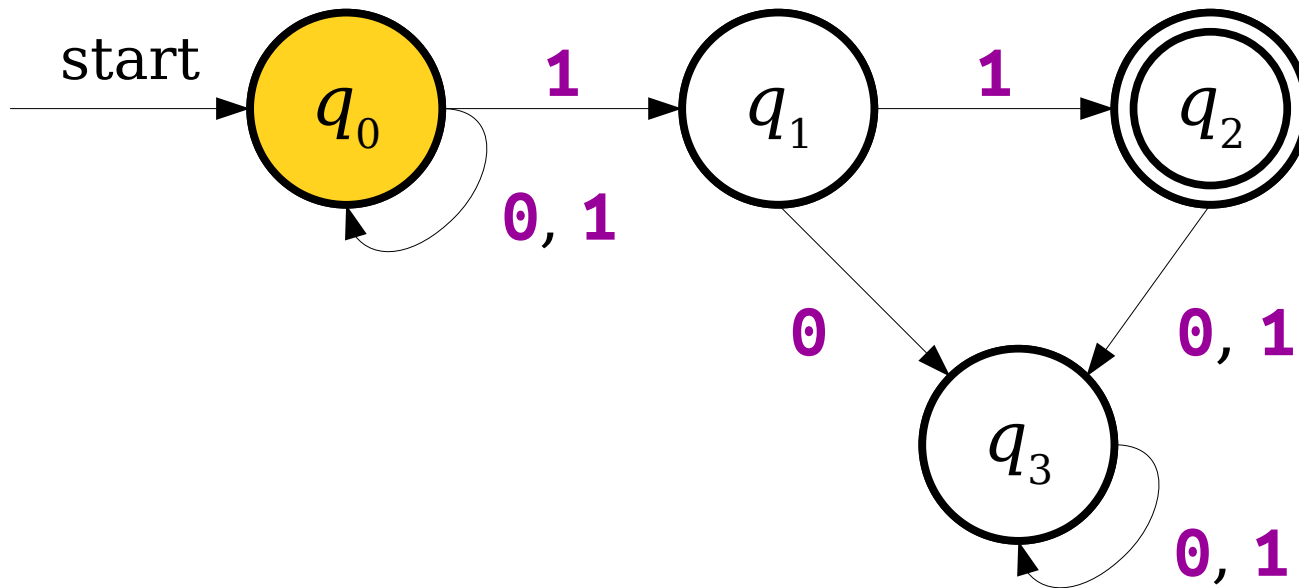
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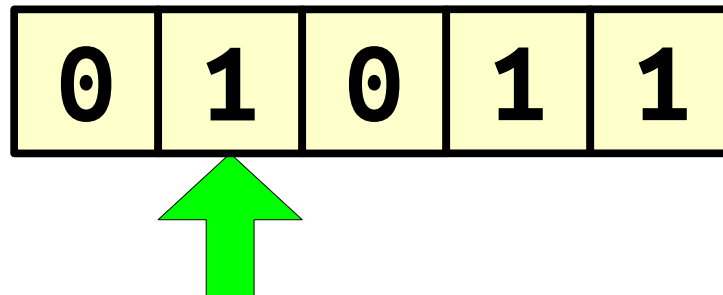
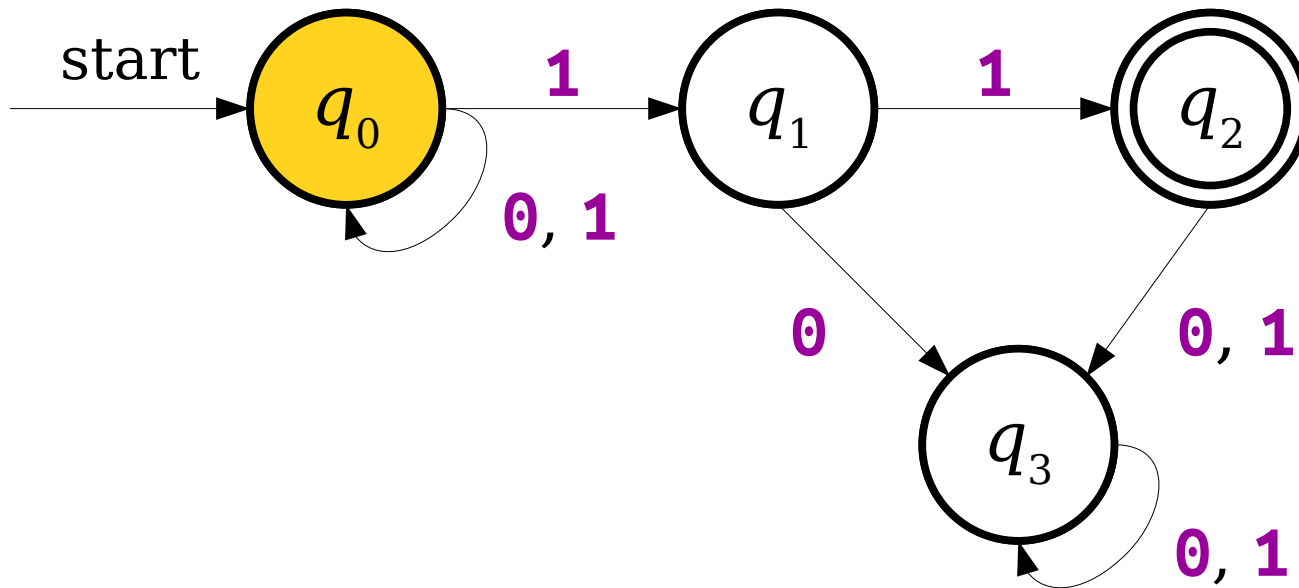
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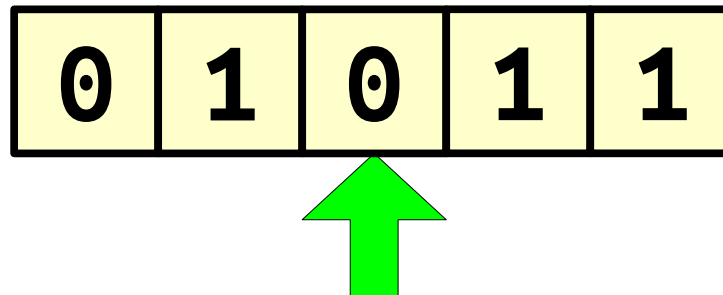
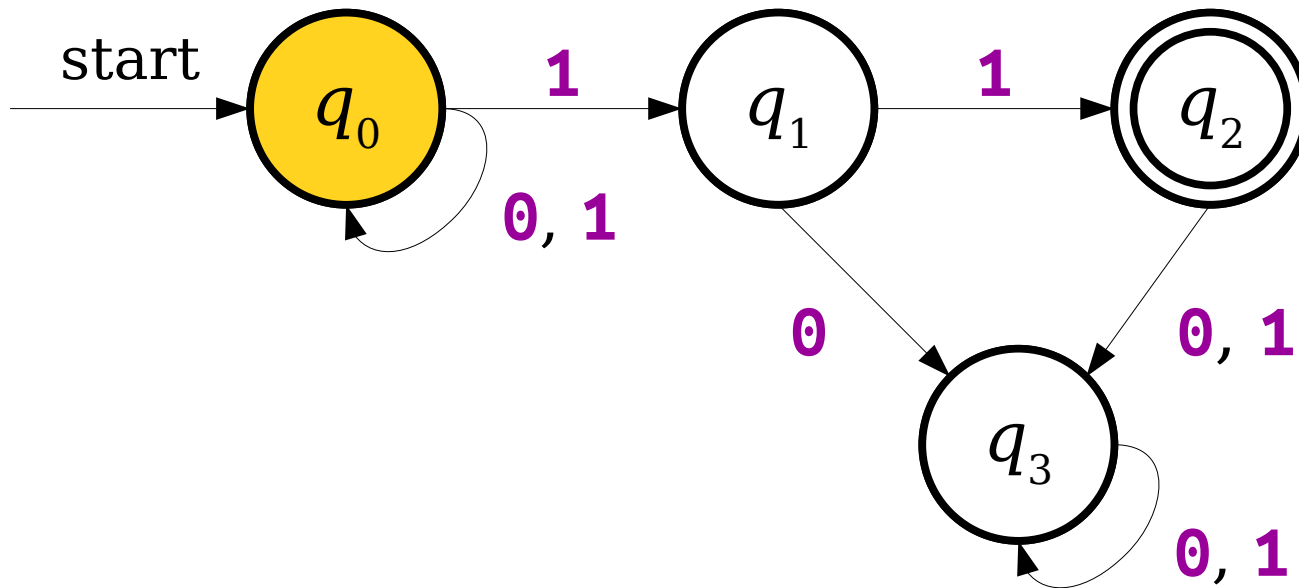
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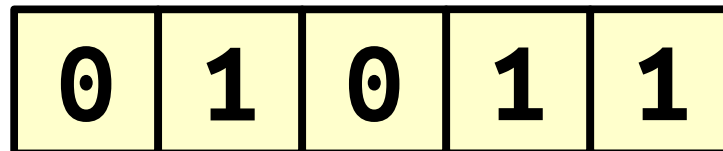
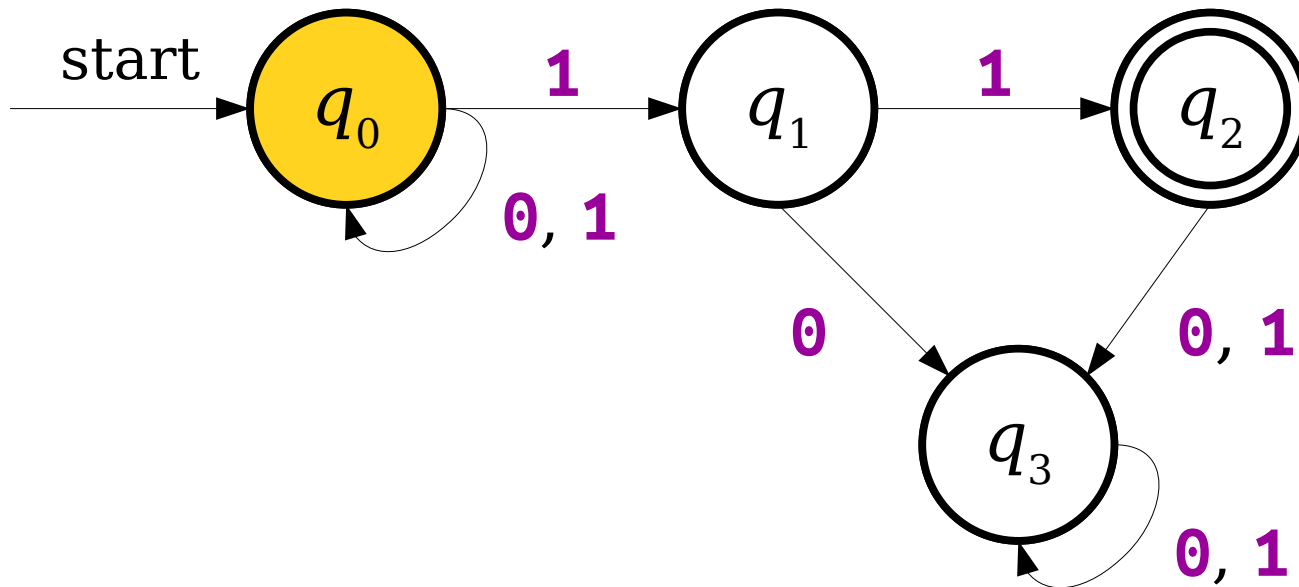
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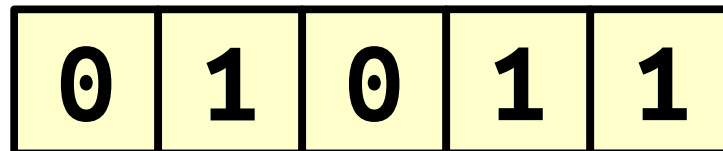
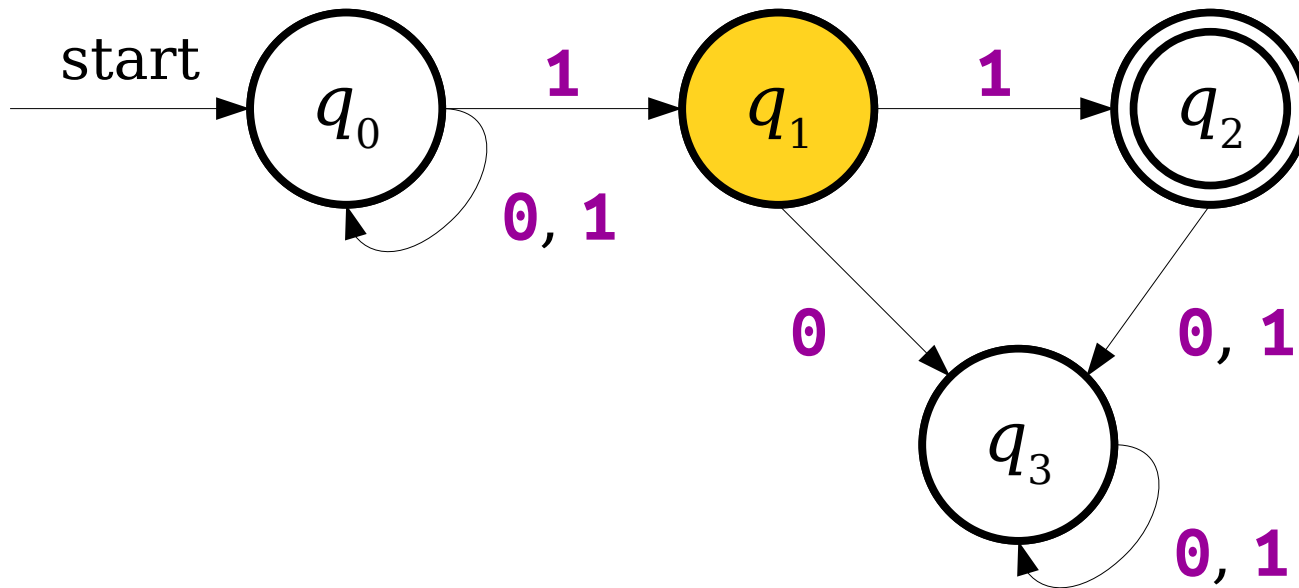
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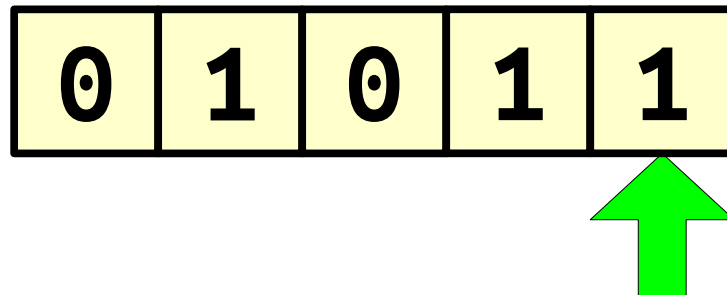
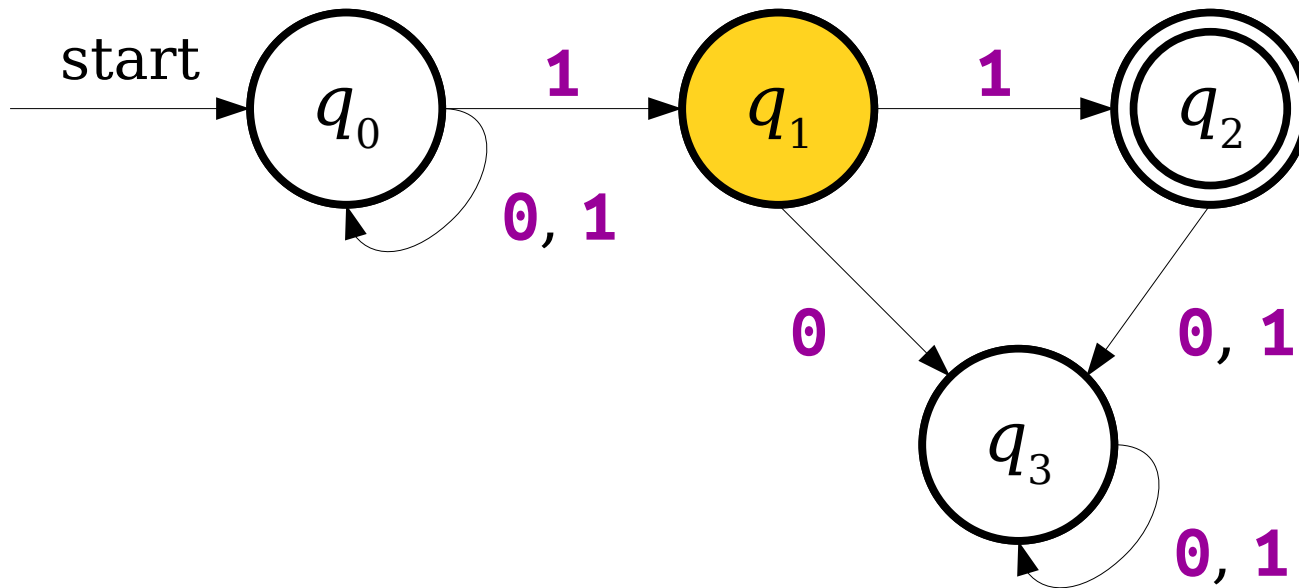
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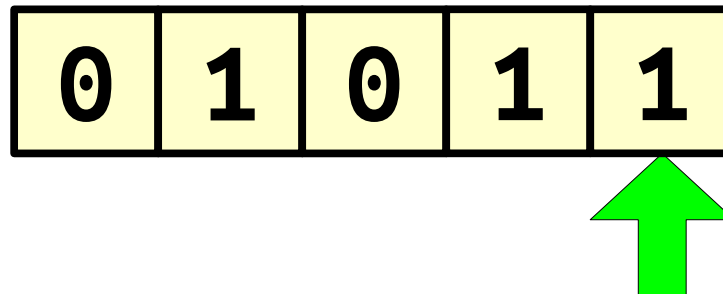
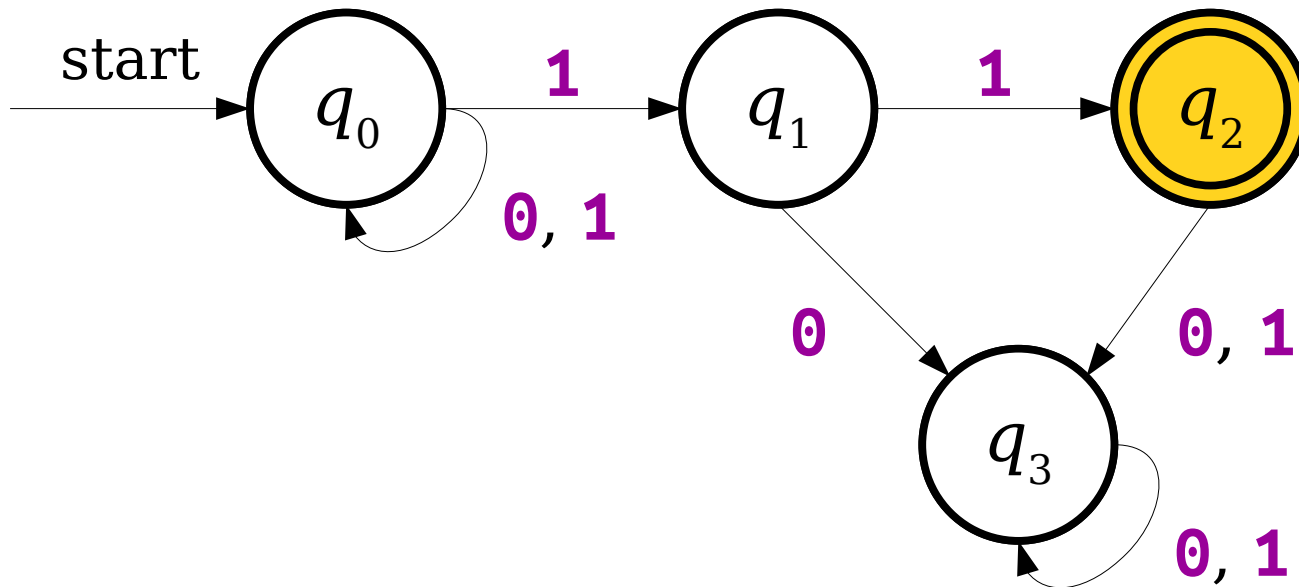
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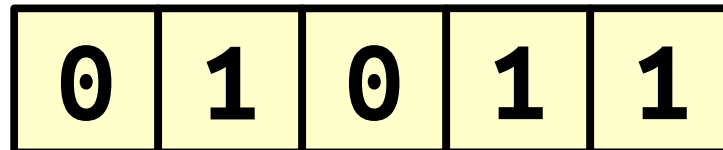
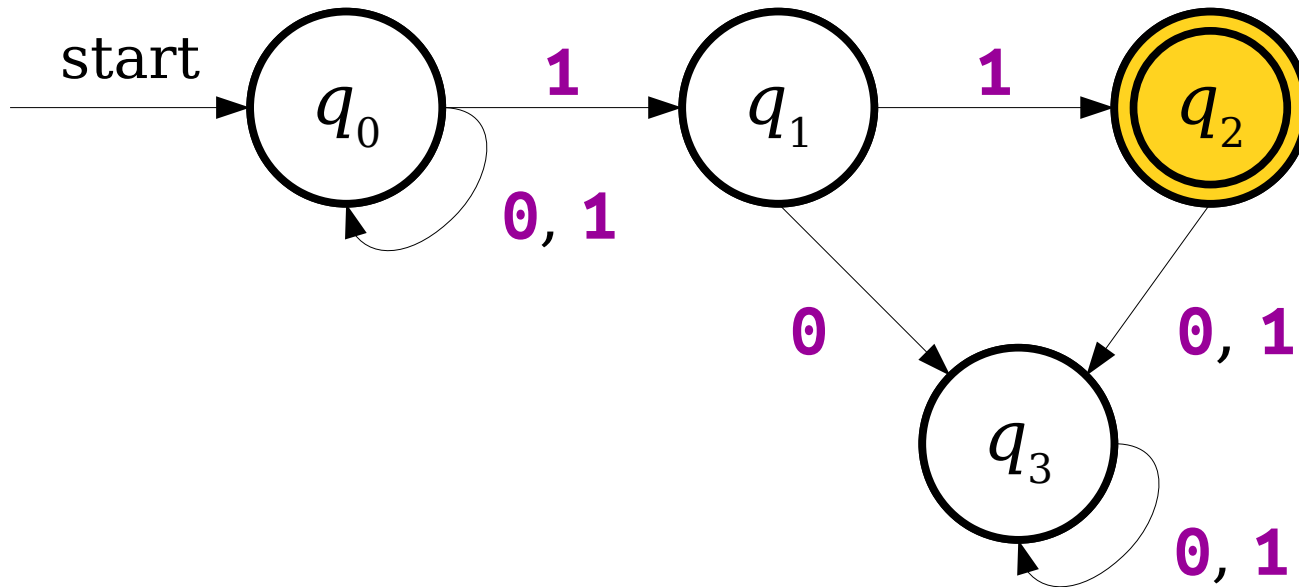
A Simple NFA



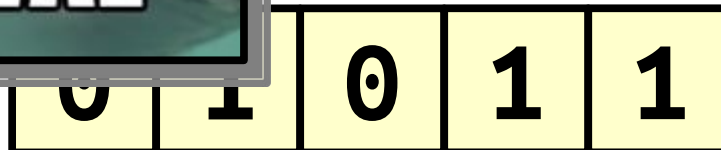
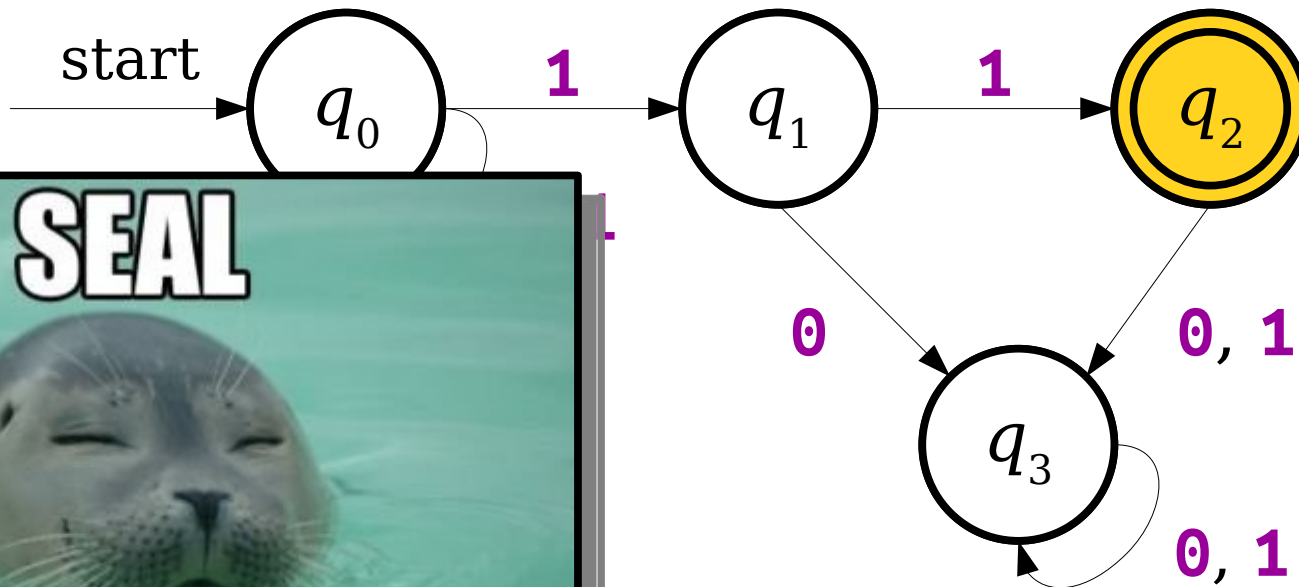
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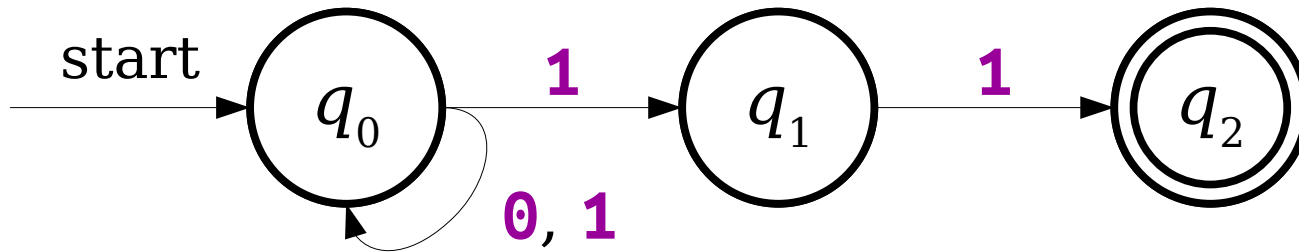
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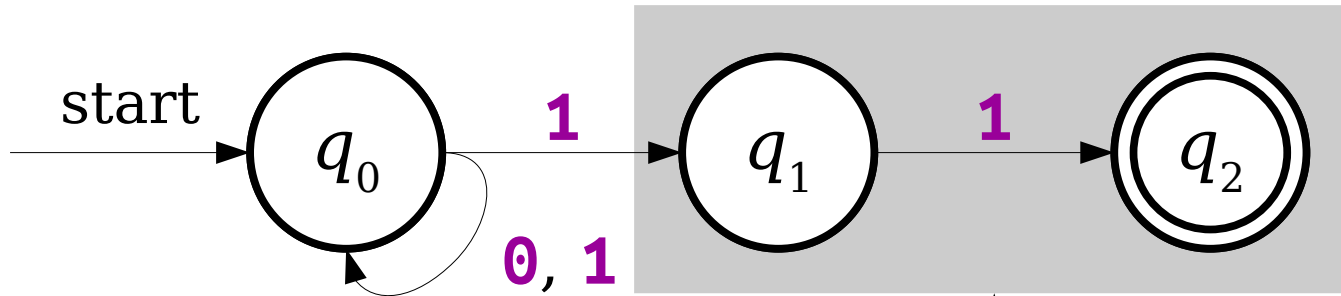
A Simple NFA



A More Complex NFA

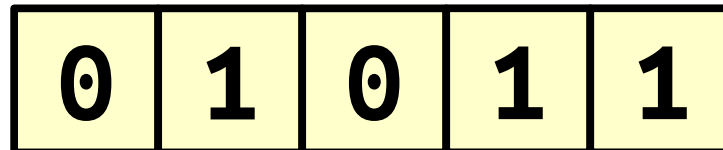
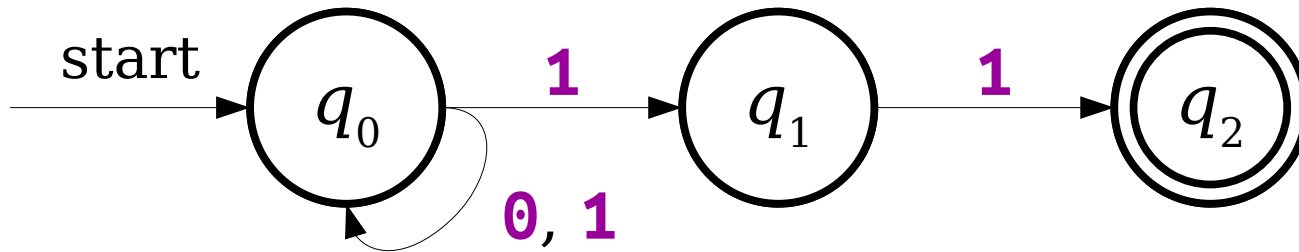


A More Complex NFA

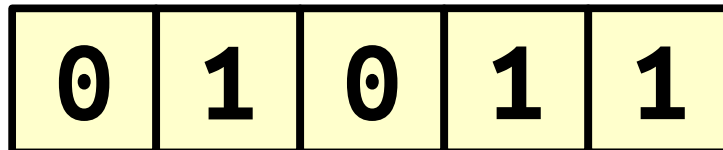
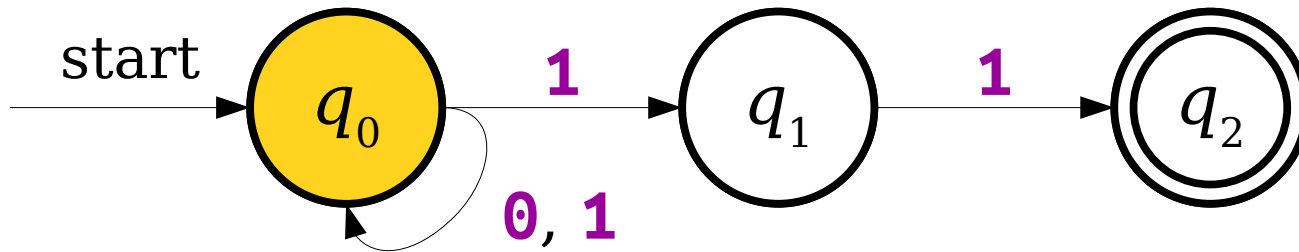


If a NFA needs to make a transition when no transition exists, the automaton **dies** and that particular path does not accept.

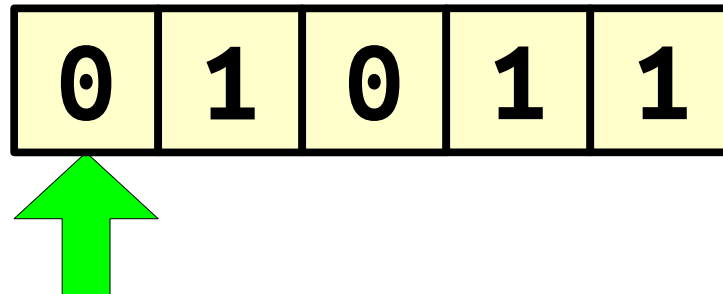
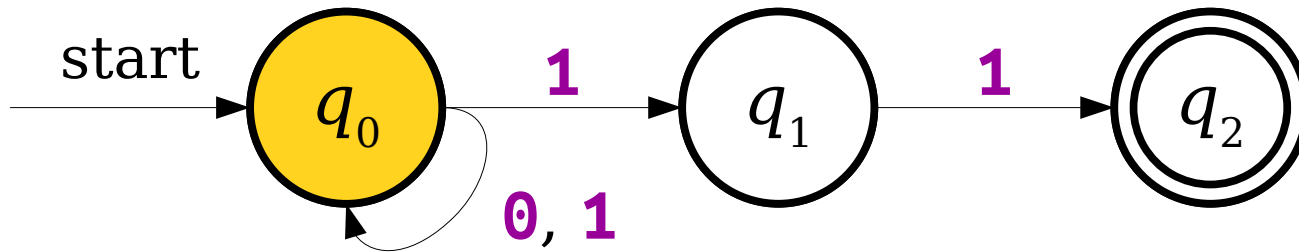
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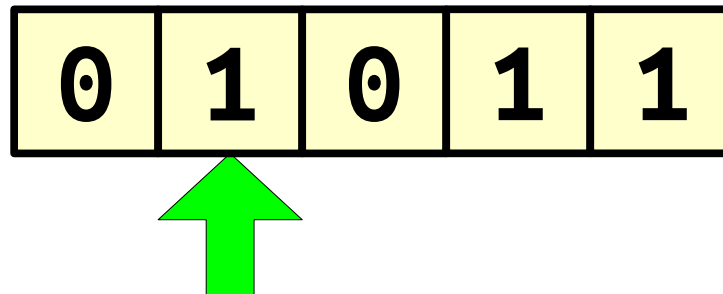
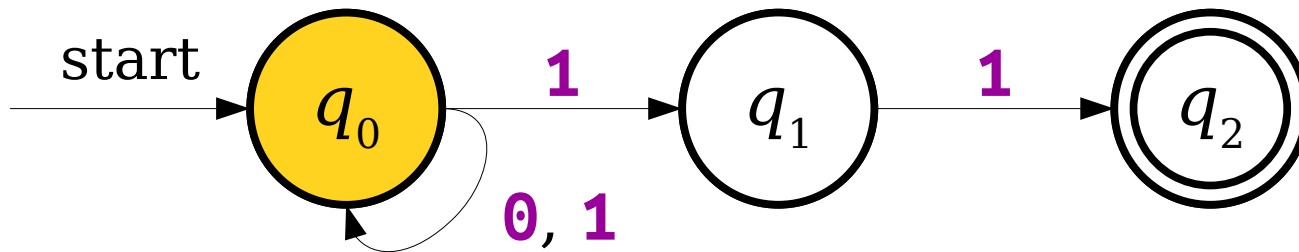
A More Complex NFA



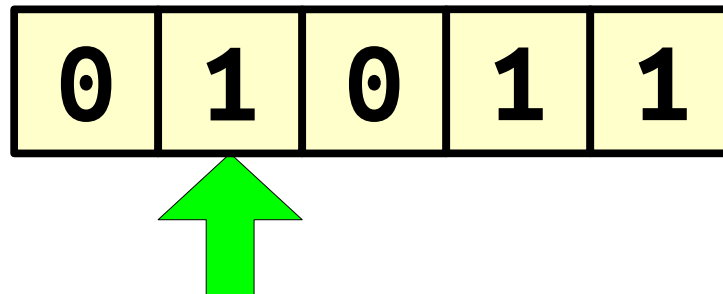
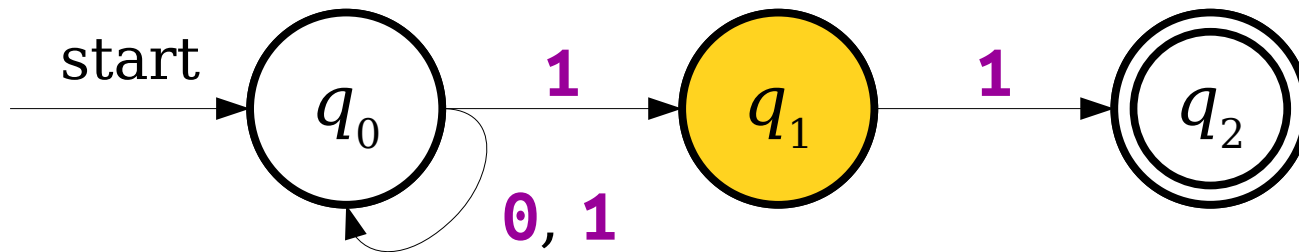
A More Complex NFA



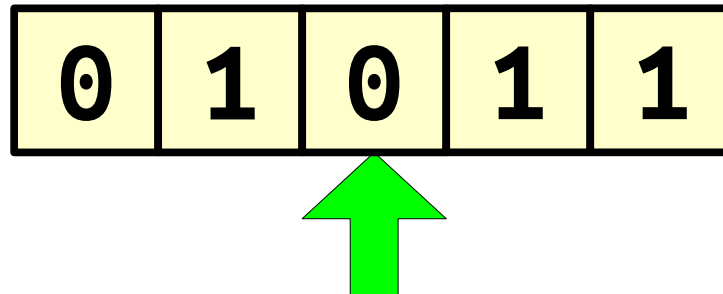
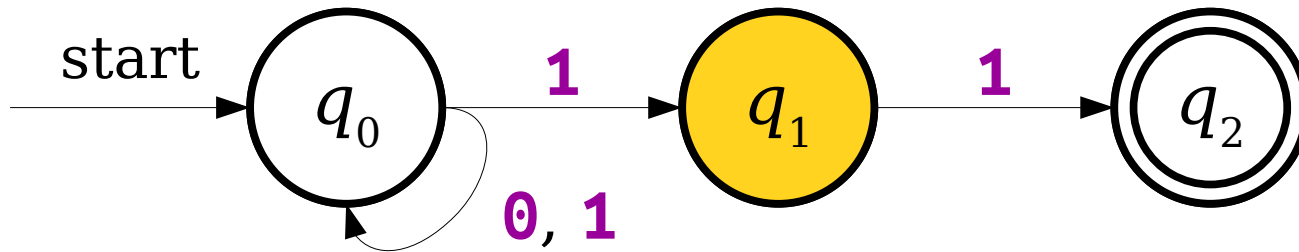
A More Complex NFA



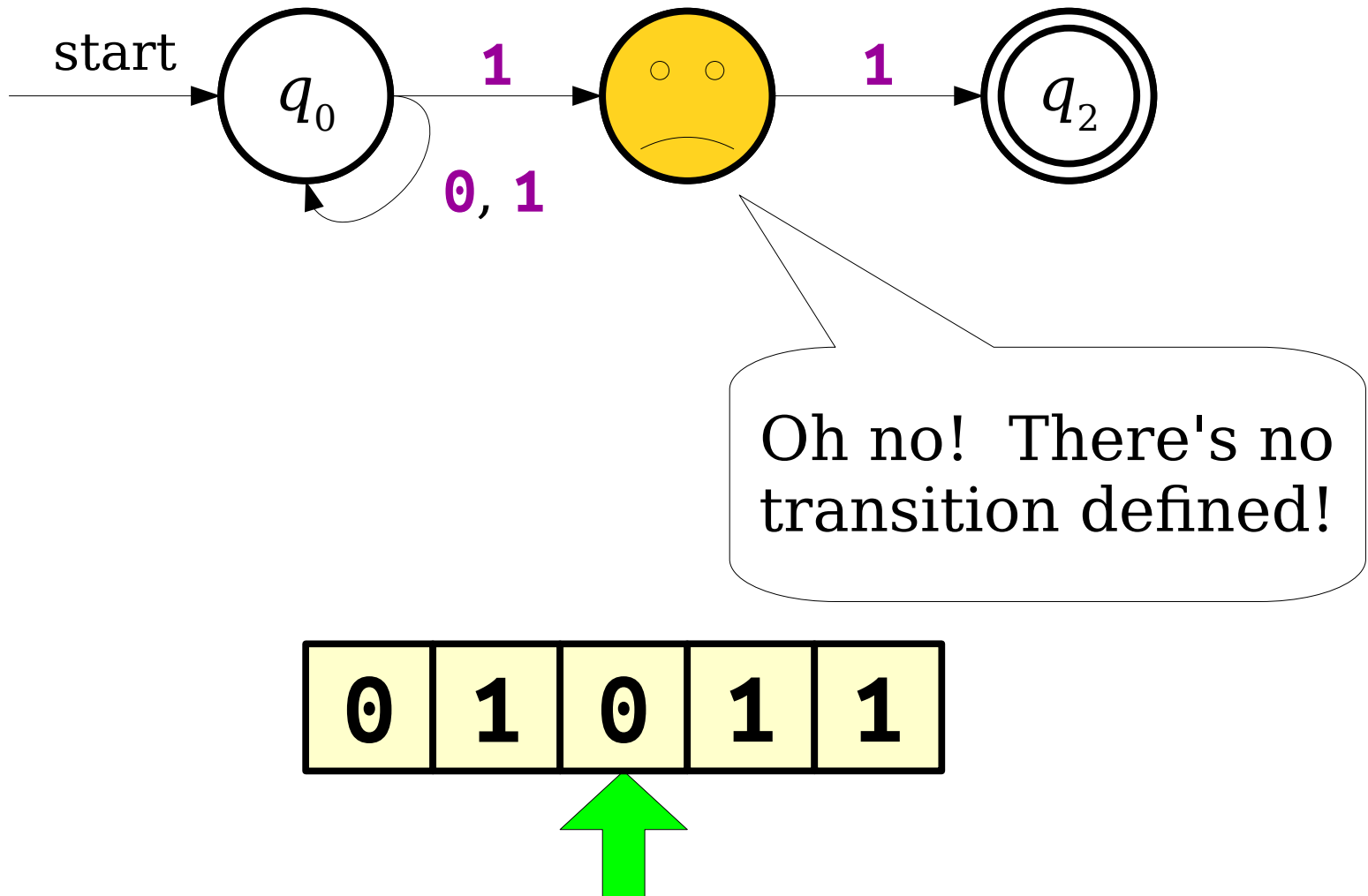
A More Complex NFA



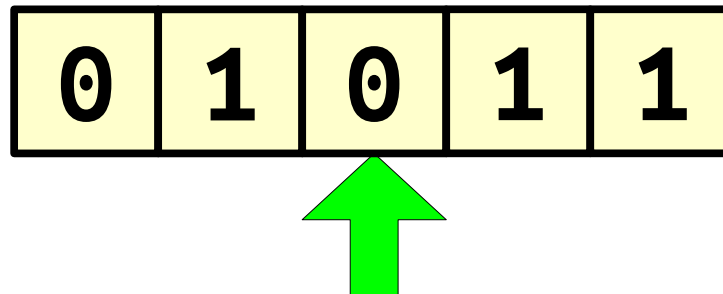
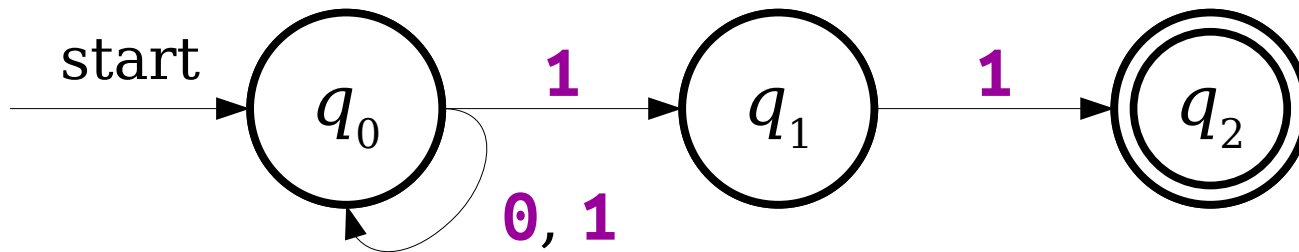
A More Complex NFA



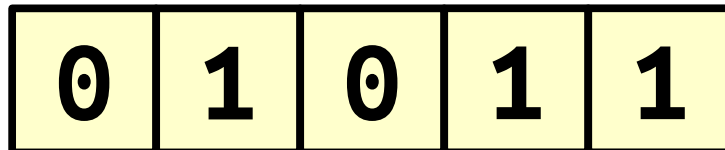
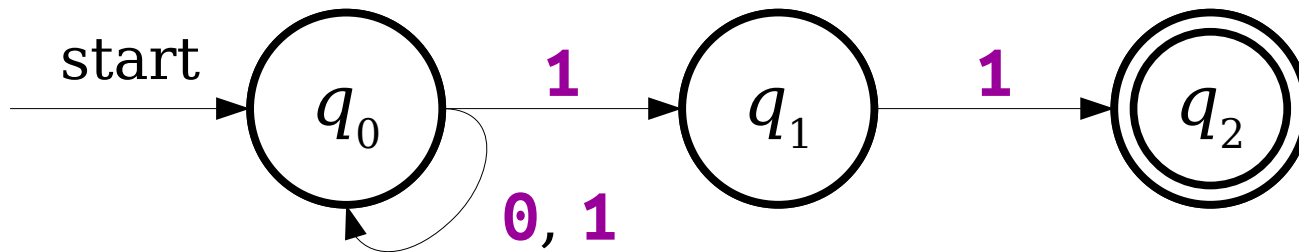
A More Complex NFA



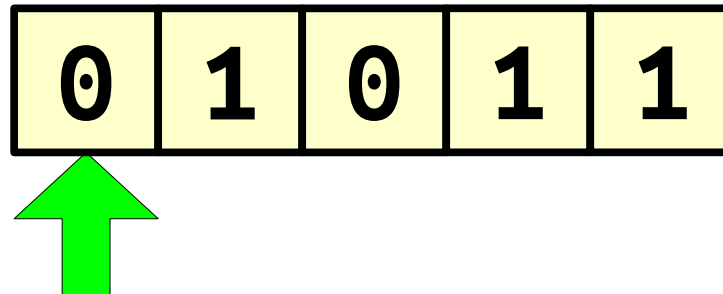
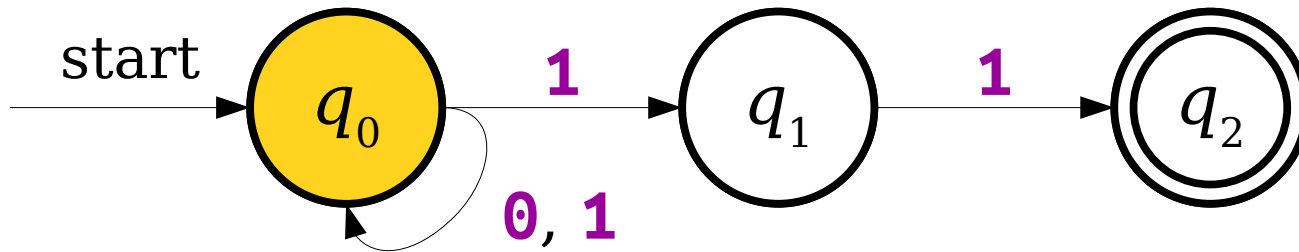
A More Complex NFA



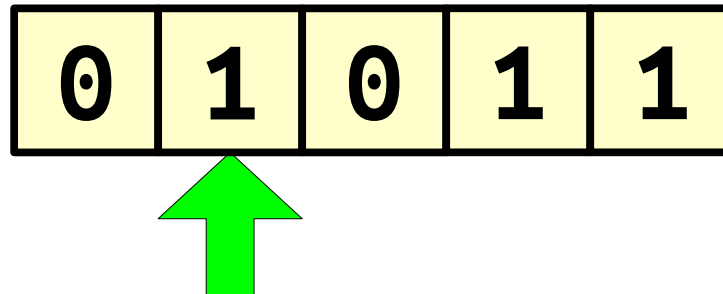
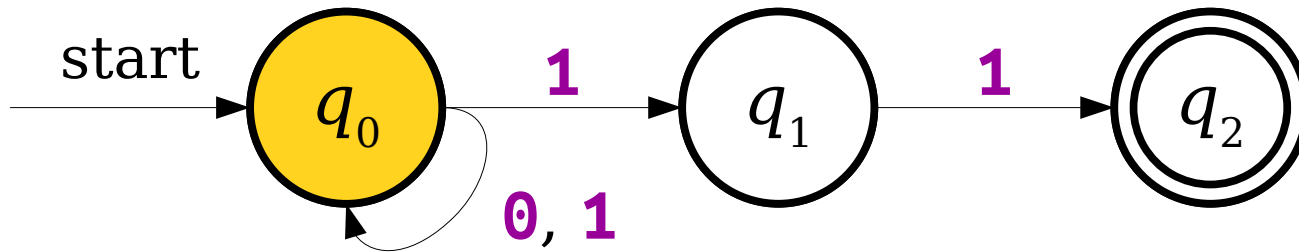
A More Complex NFA



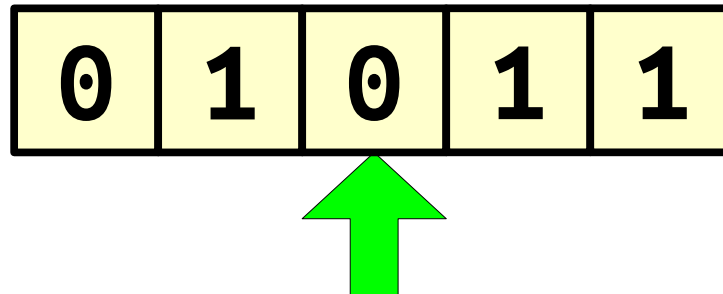
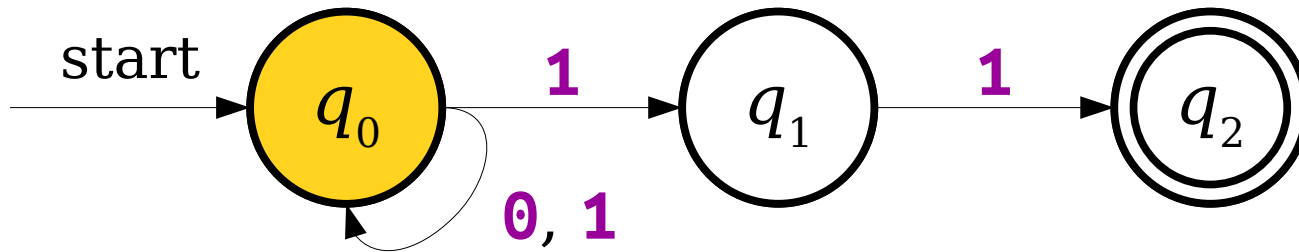
A More Complex NFA



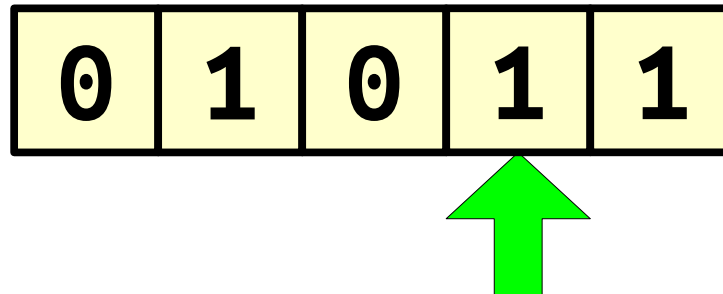
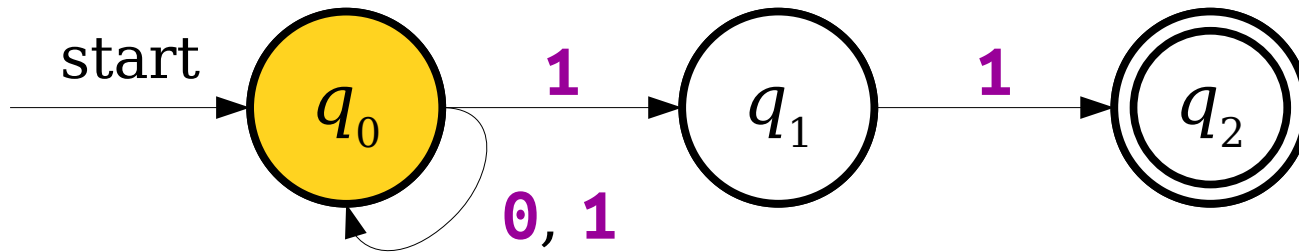
A More Complex NFA



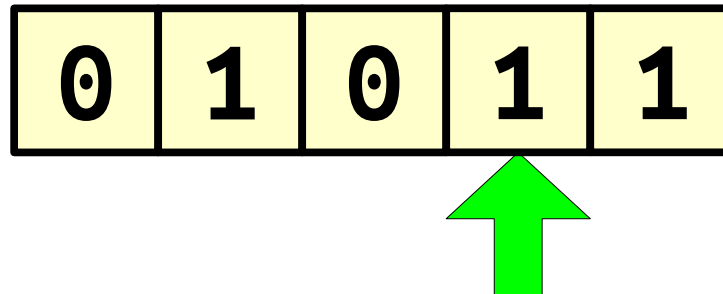
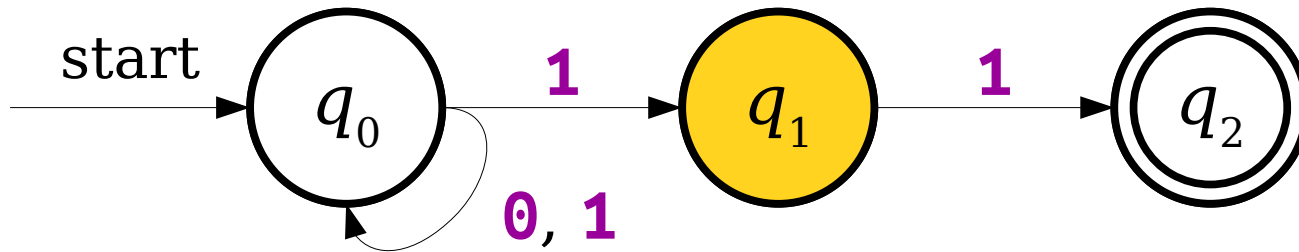
A More Complex NFA



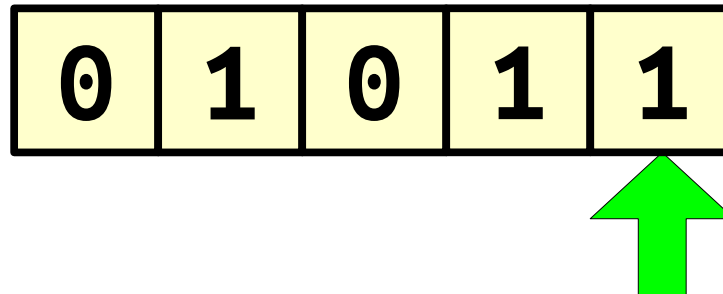
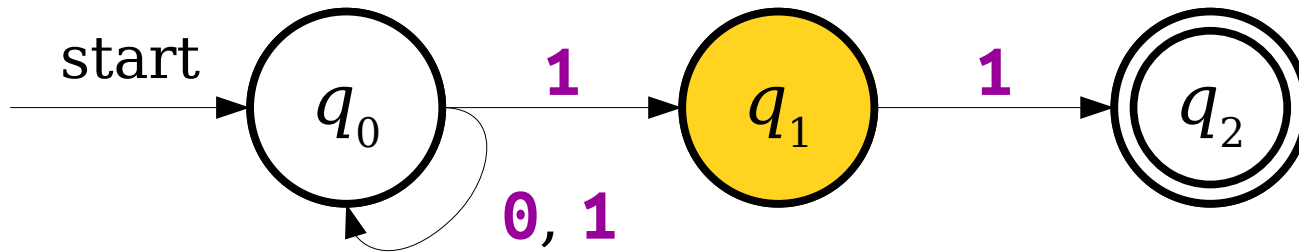
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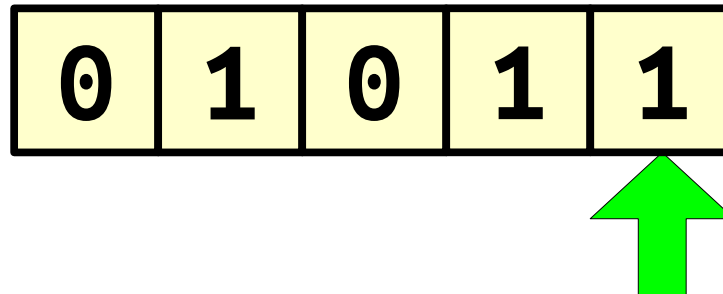
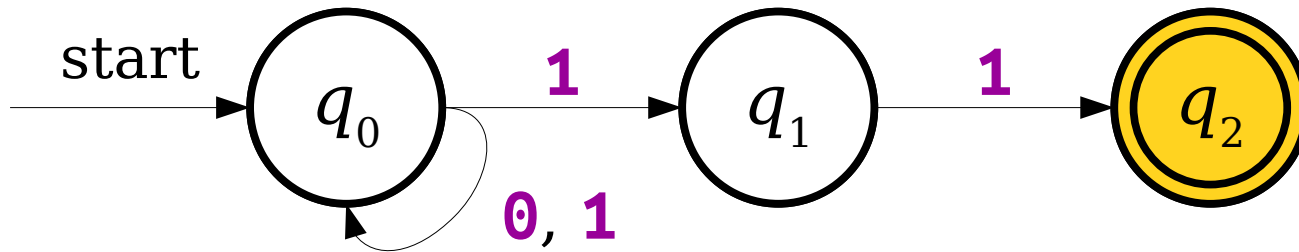
A More Complex NFA



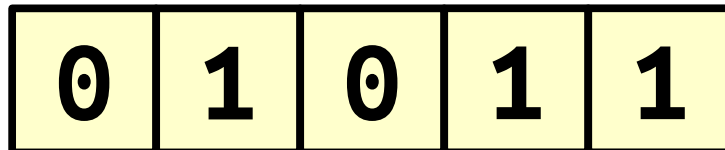
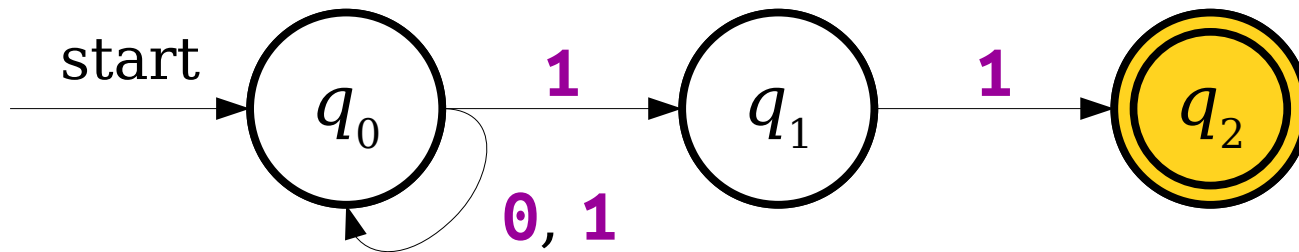
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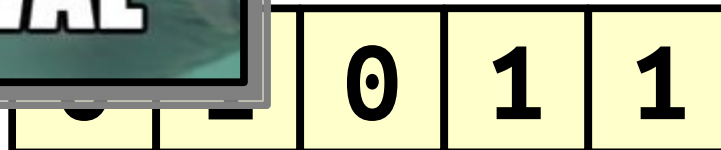
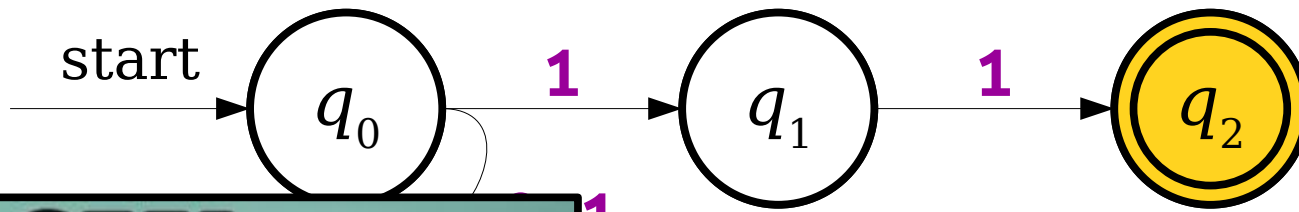
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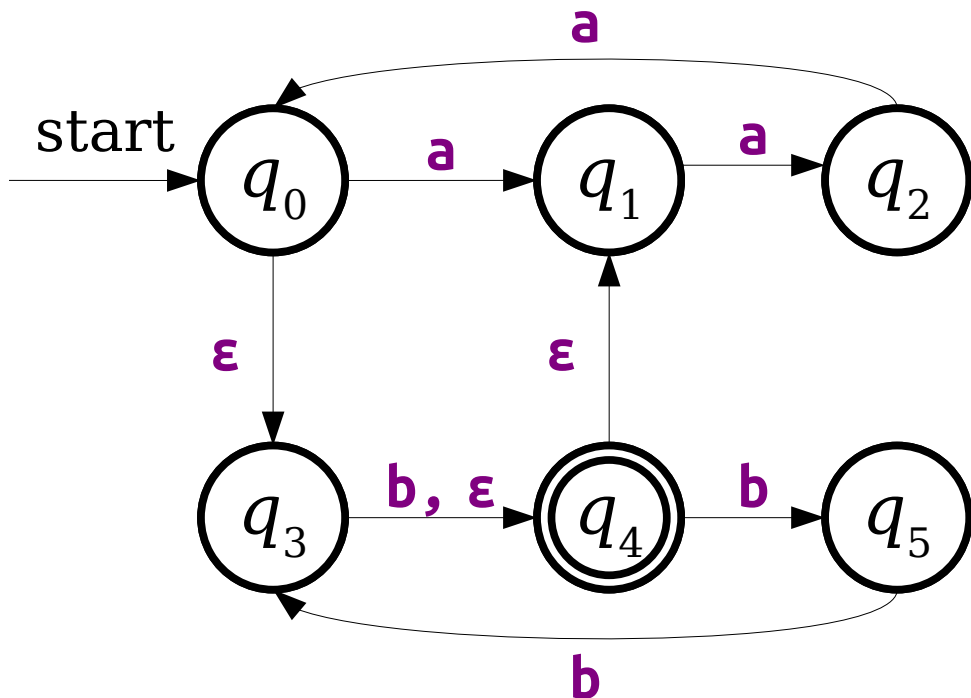


ϵ -Transitions

- NFAs have a special type of transition called the **ϵ -transition**.
- An NFA may follow any number of ϵ -transitions at any time without consuming any input.

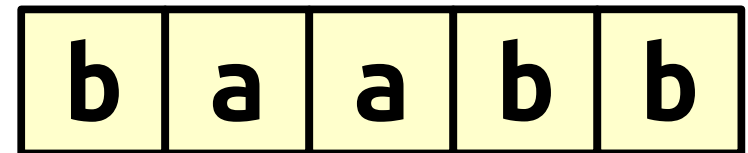
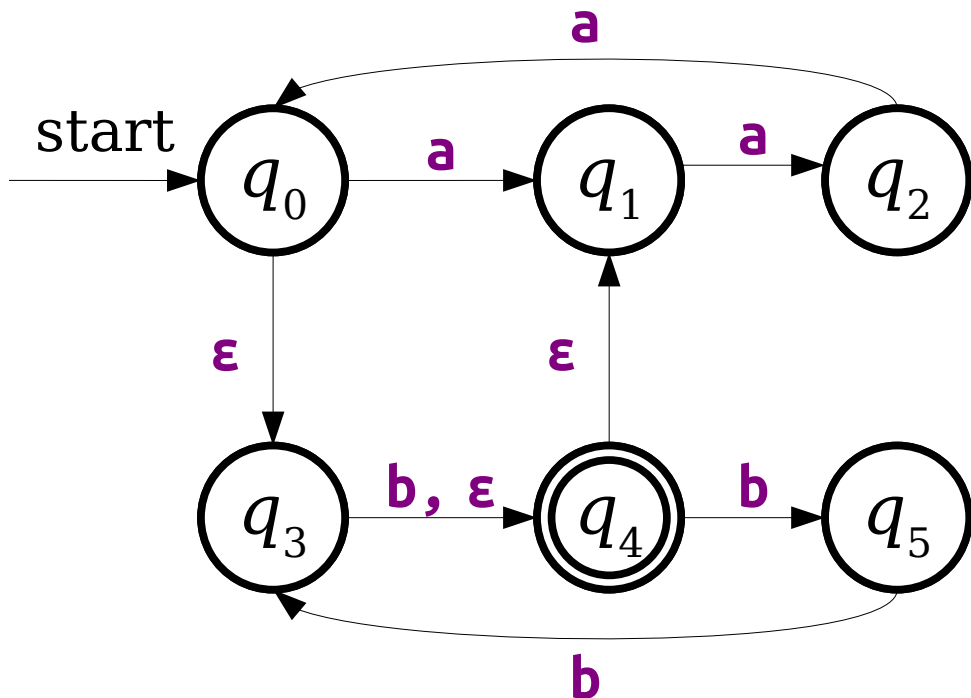
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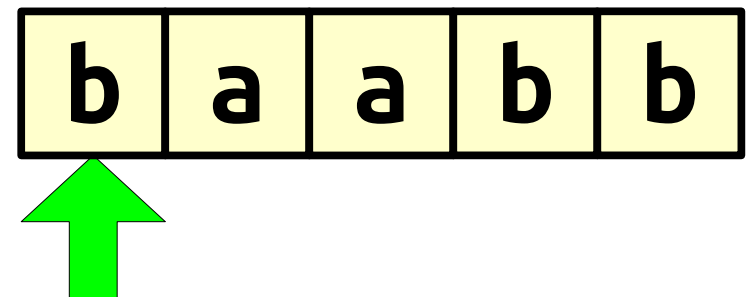
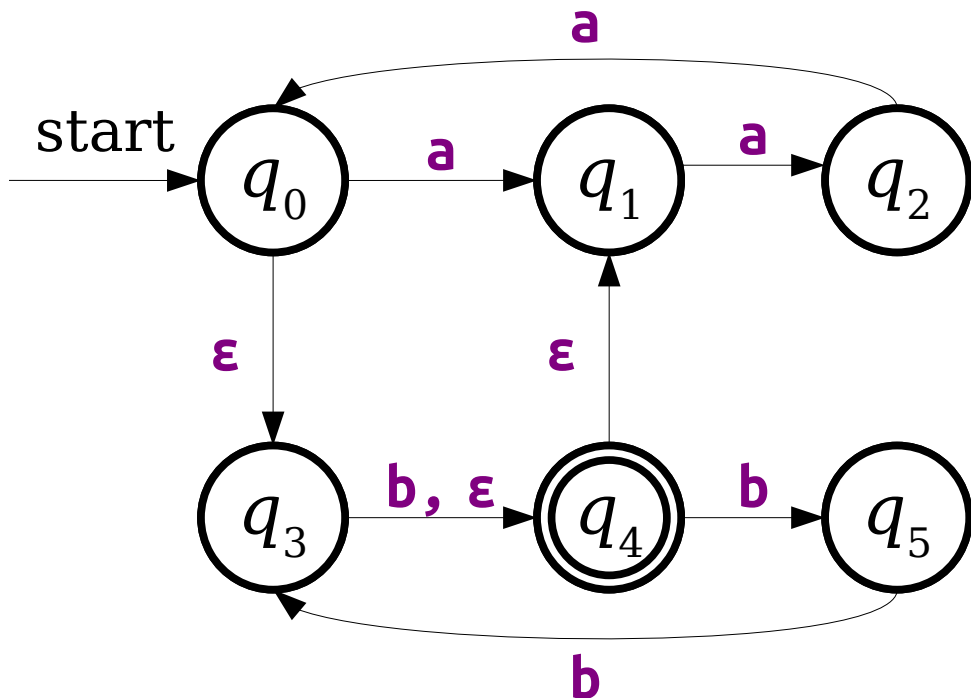
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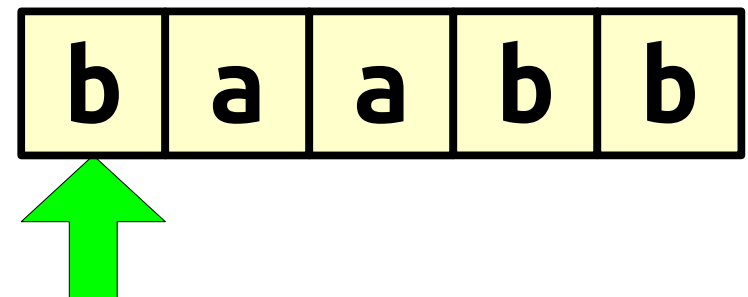
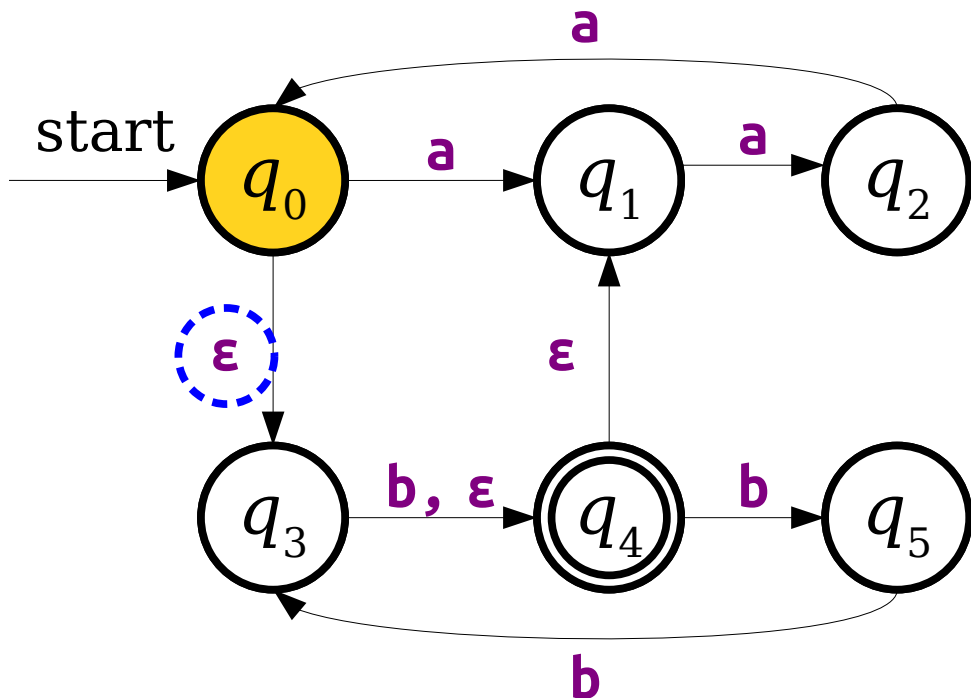
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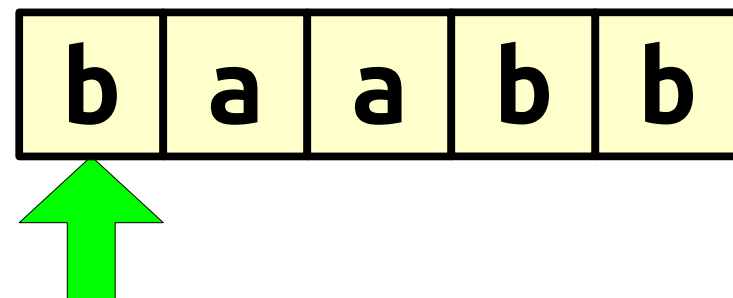
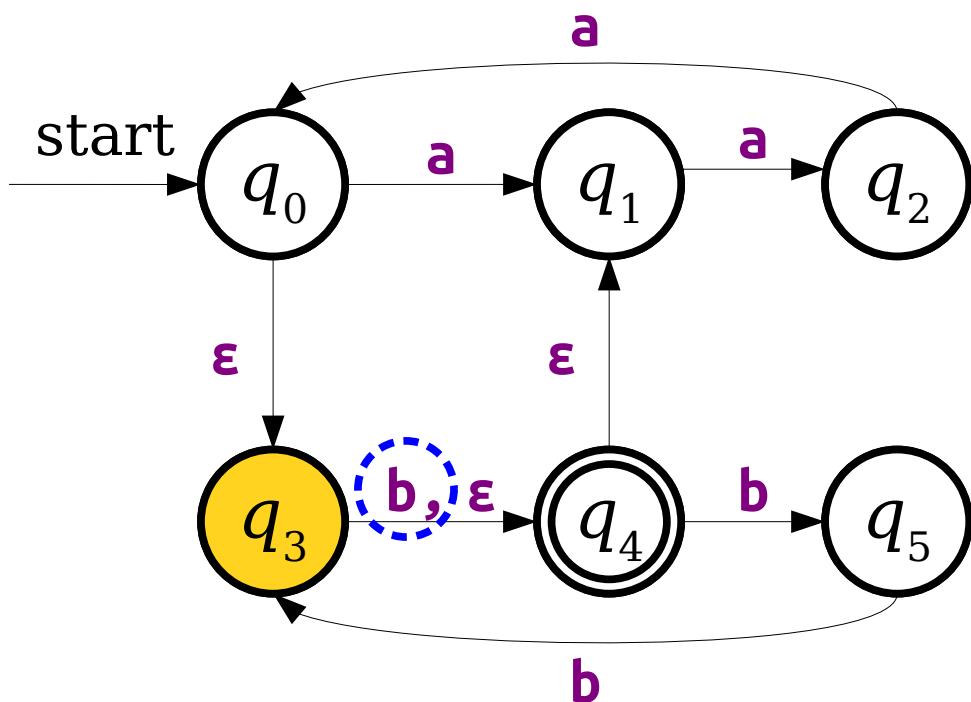
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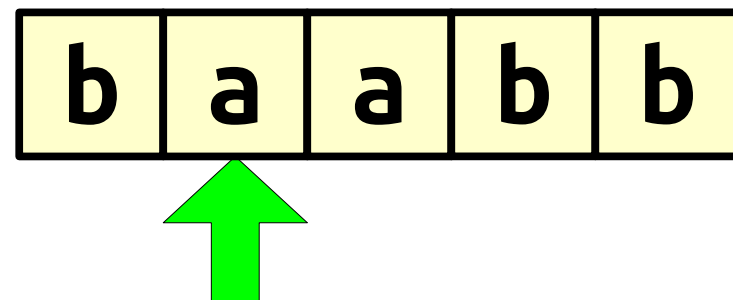
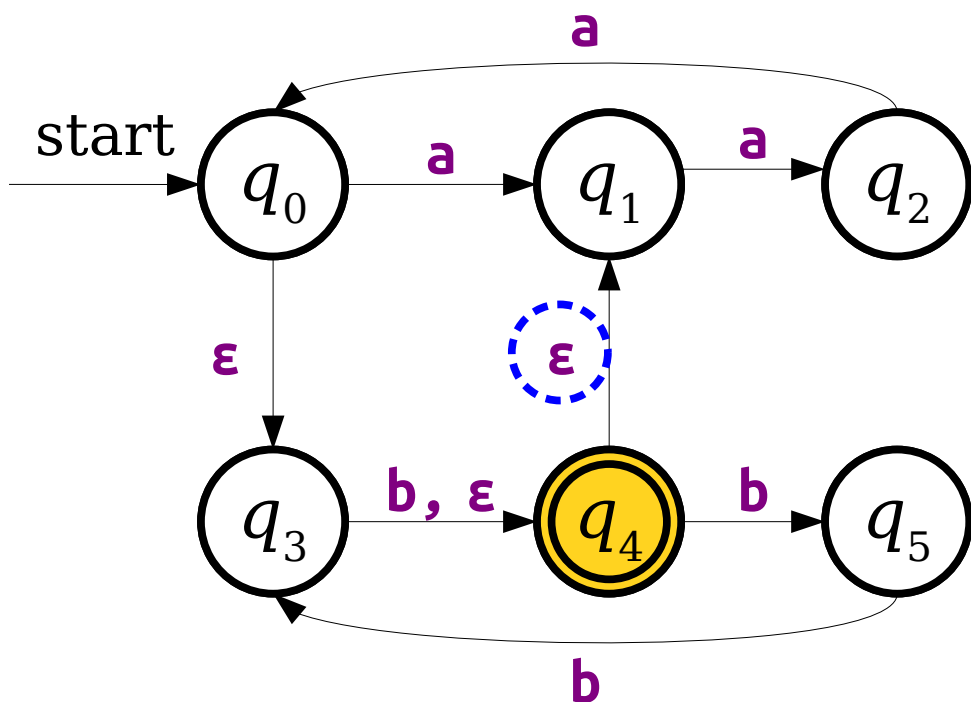
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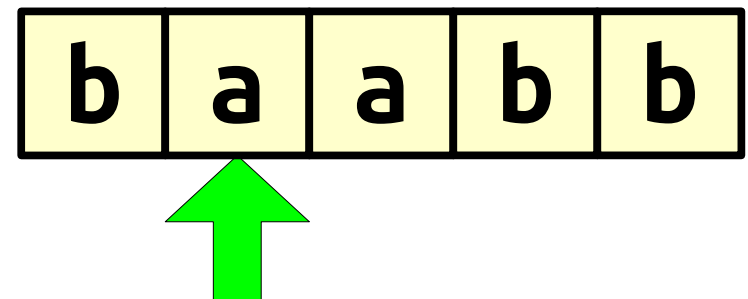
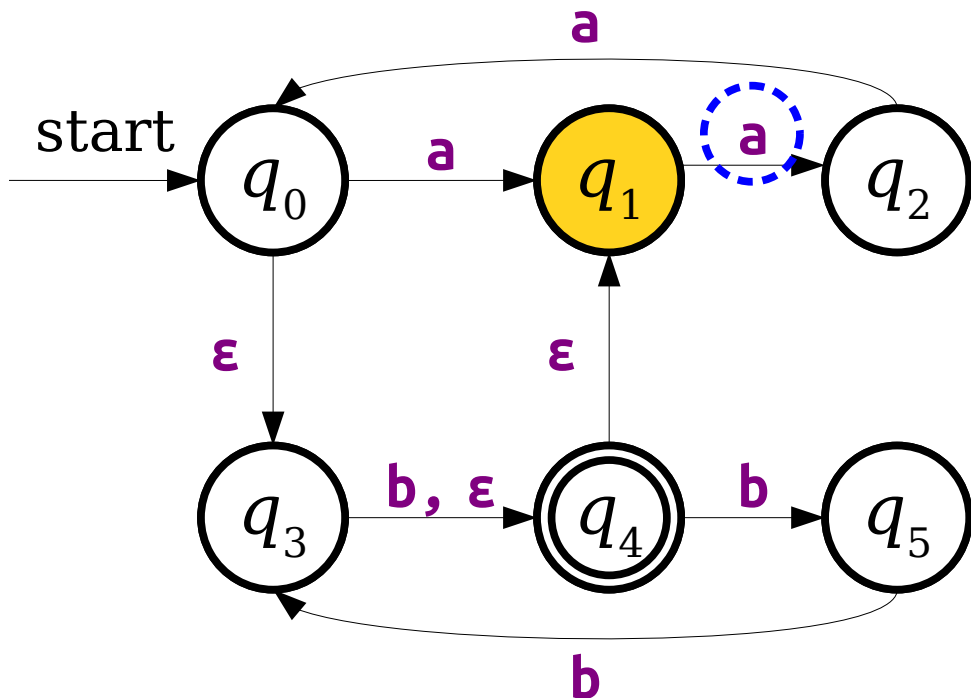
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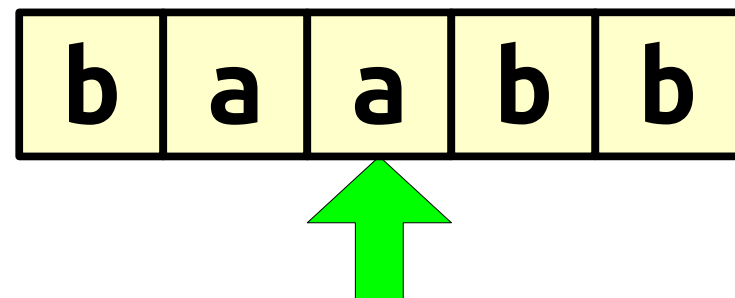
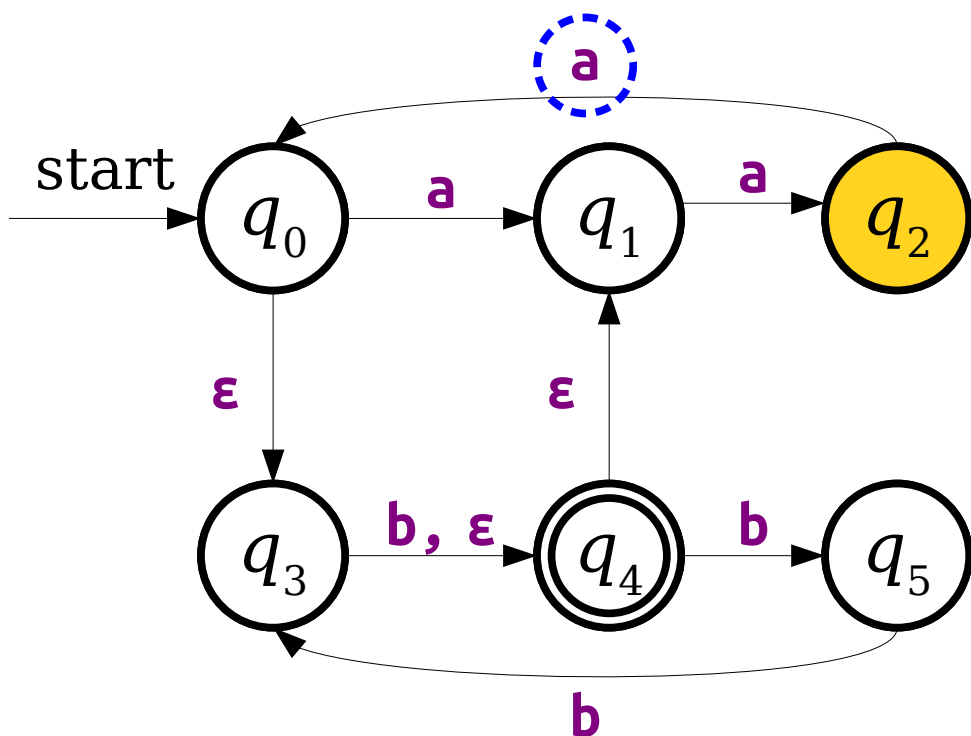
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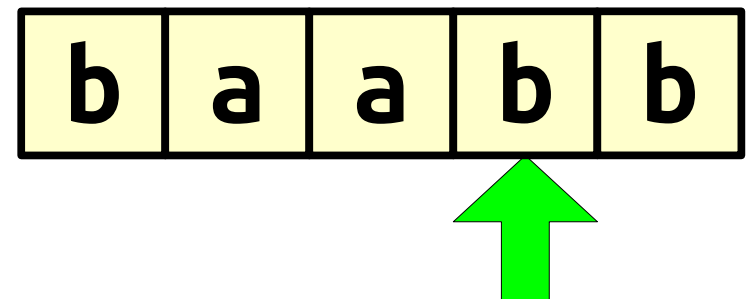
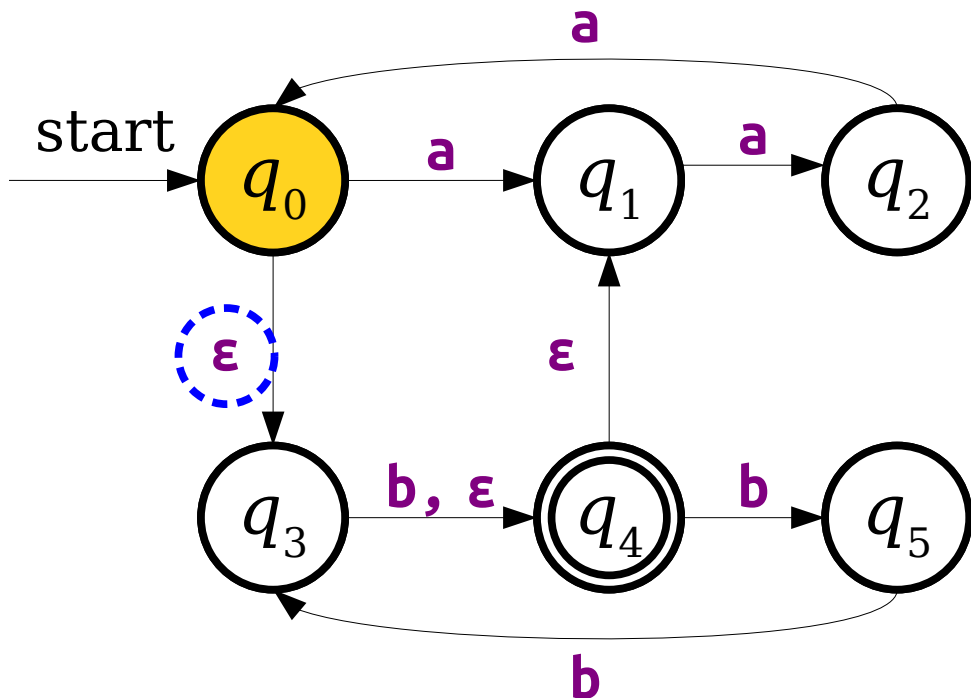
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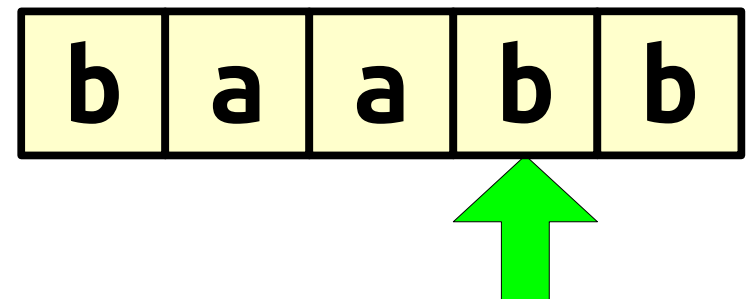
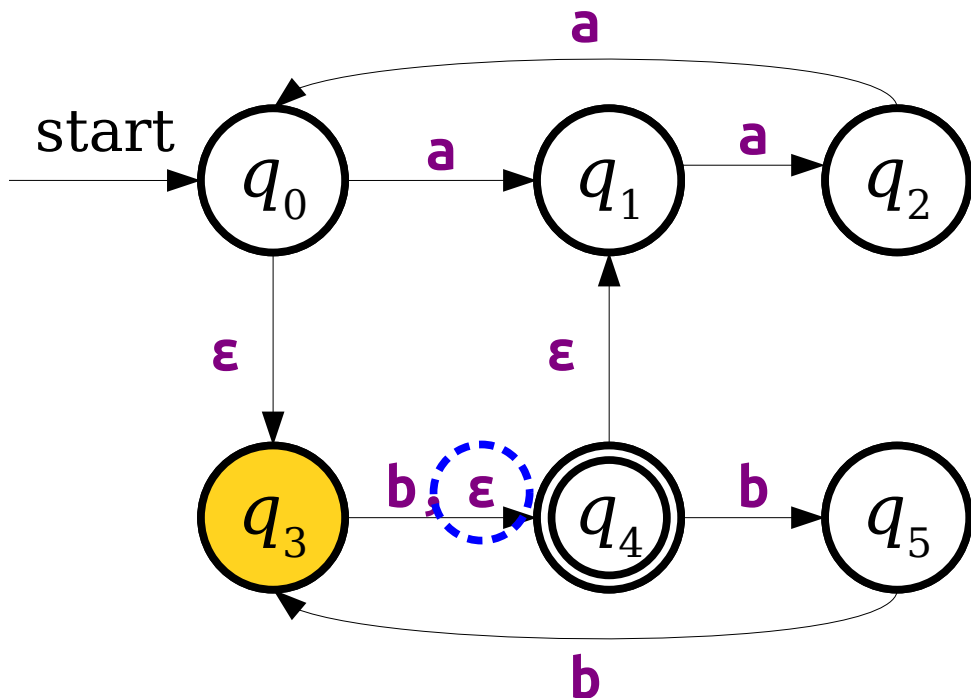
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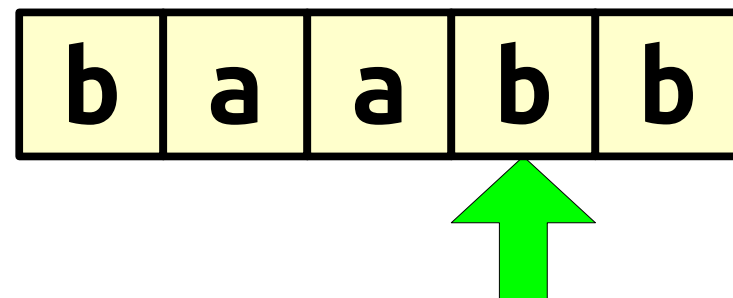
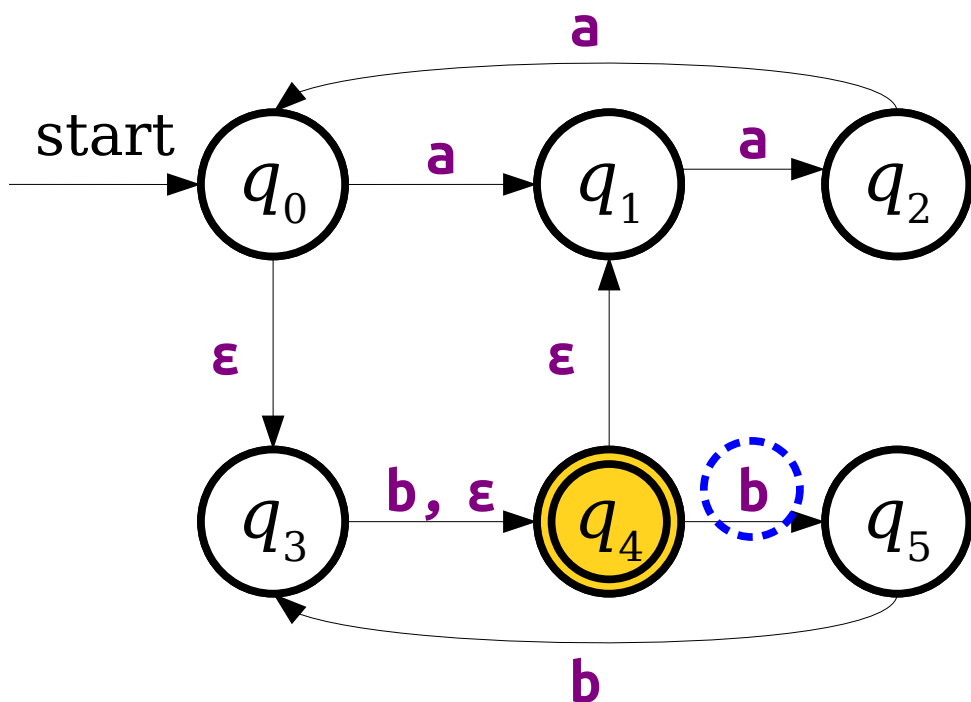
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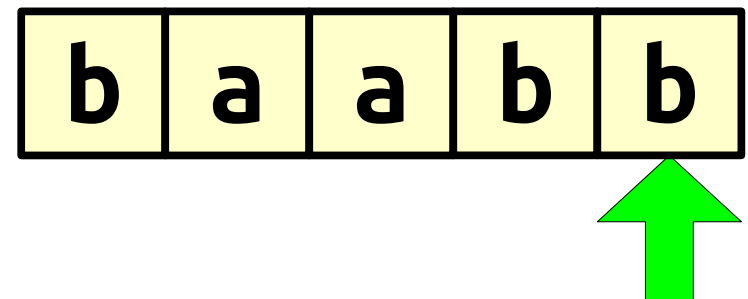
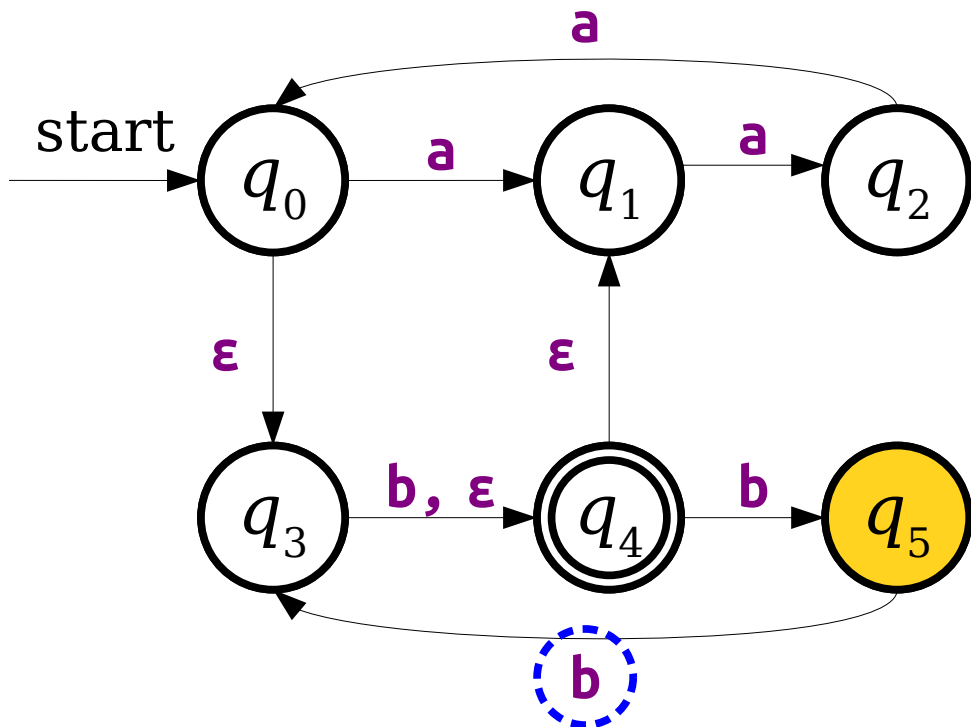
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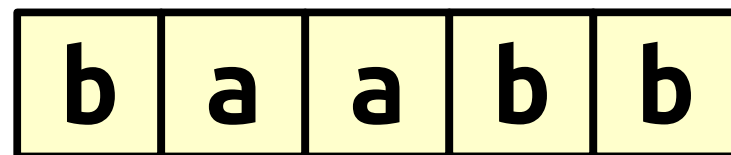
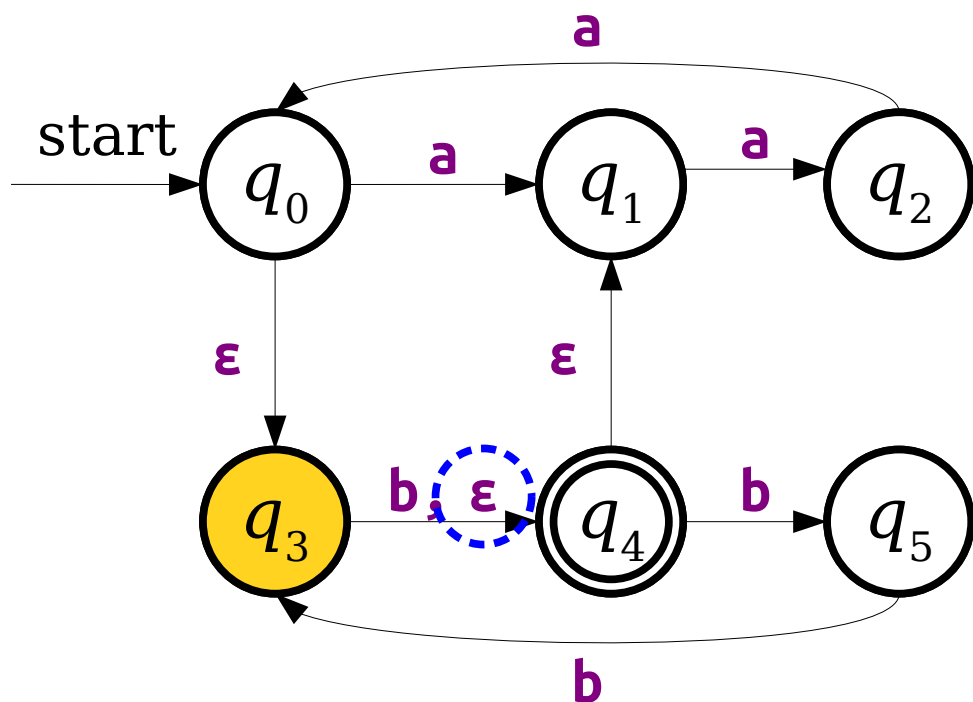
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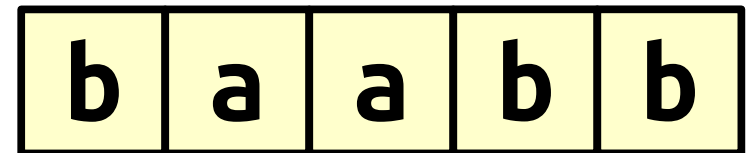
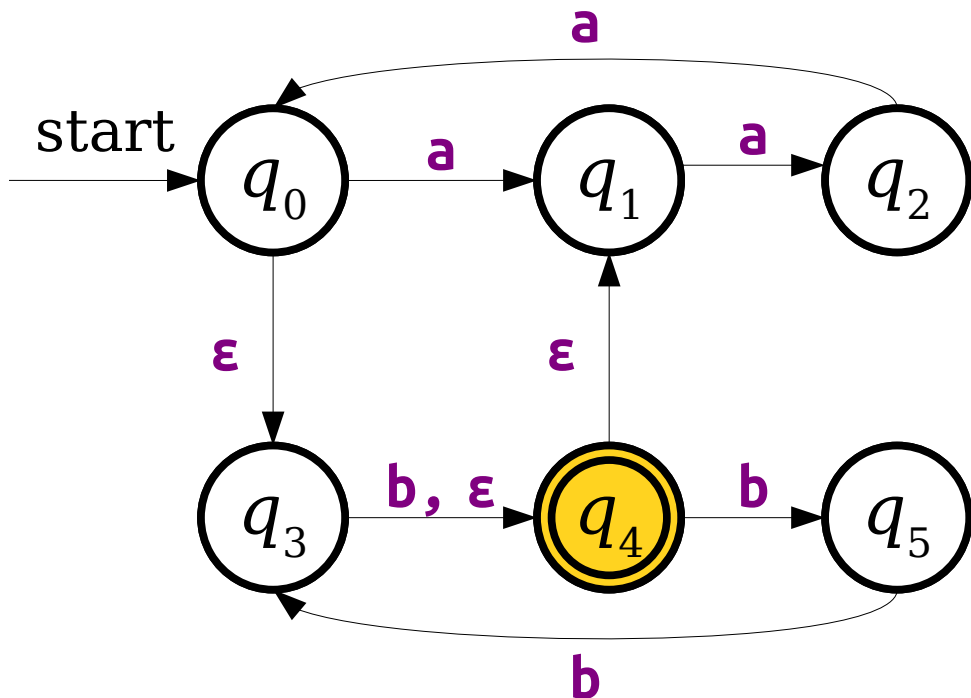
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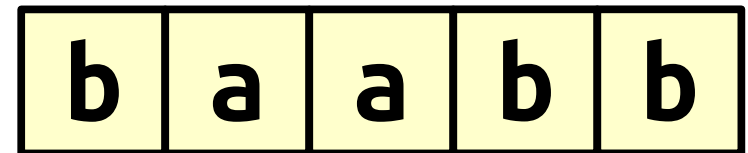
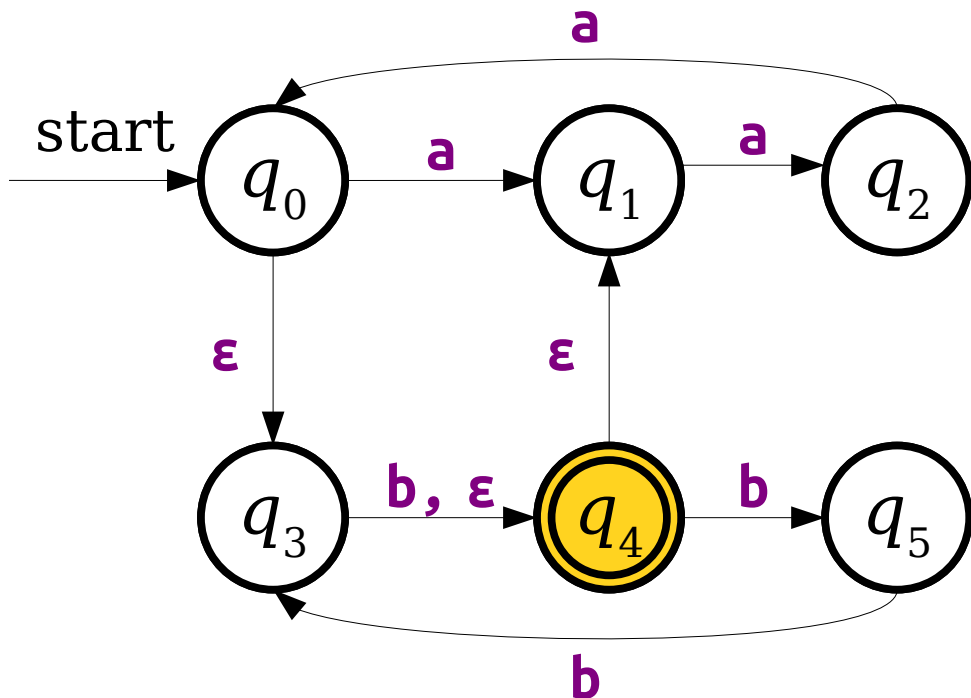
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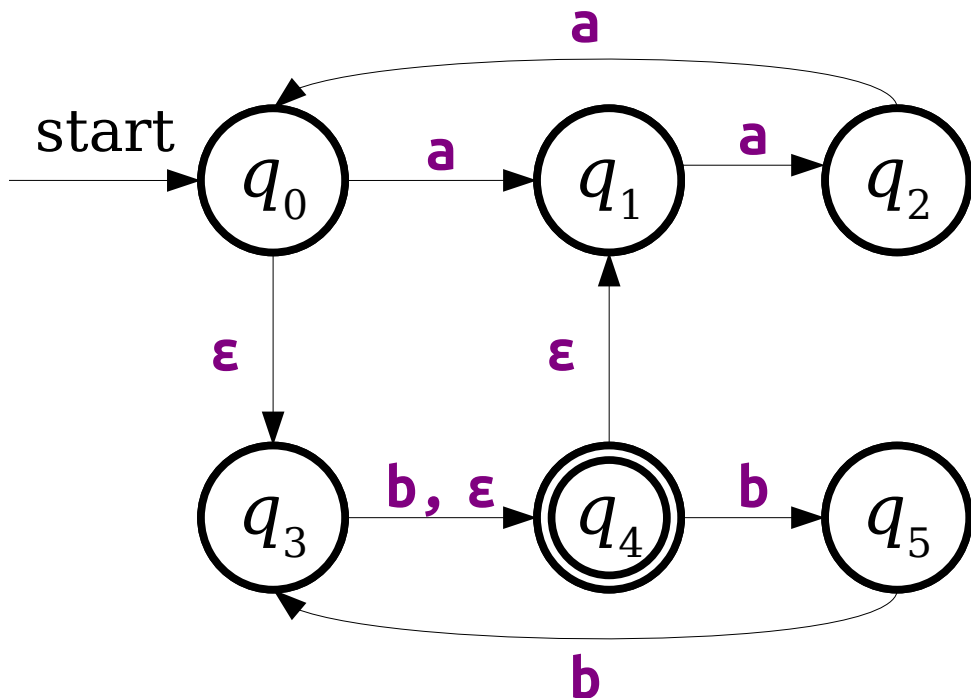
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Not at all fun or
rewarding exercise:
what is the language of
this NFA?

ϵ -Transitions

- NFAs have a special type of transition called the **ϵ -transition**.
- An NFA may follow any number of ϵ -transitions at any time without consuming any input.
- NFAs are not *required* to follow ϵ -transitions. It's simply another option at the machine's disposal.

NFAs

- An NFA is defined relative to some alphabet Σ .
- For each state in the NFA, there may be *any number of* transitions defined for each symbol in Σ , plus any number of ϵ -transitions.
 - This is the “nondeterministic” part of NFA.
- There is a unique start state.
- There are zero or more accepting states.

DFA_s

- A DFA is defined relative to some alphabet Σ .
- For each state in the DFA, there must be *exactly one* transition defined for each symbol in Σ . Additionally, ϵ -transitions are not allowed.
 - This is the “deterministic” part of DFA.
- There is a unique start state.
- There are zero or more accepting states.

Designing NFAs

Designing NFAs

- ***Embrace the nondeterminism!***
- Good model: ***Guess-and-check***:
 - Is there some information that you'd really like to have? Have the machine *nondeterministically guess* that information.
 - Then, have the machine *deterministically check* that the choice was correct.
- The *guess* phase corresponds to trying lots of different options.
- The *check* phase corresponds to filtering out bad guesses or wrong options.